Projection, Measure, and Idempotent Relations: Independent Axioms and a Fixed-Point Coupling Law

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Abstract

We introduce a minimal ZFC–internal axiom system for *pre-structural data*

$$(X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta),$$

where $\Pi_R : X \to R$ is a designated map and $G \subseteq X \times X$ is a measurable relation. Those pre-structural data satisfying Axioms I–III are called *admissible structural models*; the axioms couple a finitely additive measure, an idempotent retraction, and an idempotent symmetric relation through a single coupling law (Axiom III). The main results are the following. Consistency: the axiom system is satisfiable in ZFC, with explicit finite and countable models, including finite families for which $\eta \neq 0$. Independence: the three axioms, and the three subclauses of Axiom III (conservation, projection–measure invariance, coupling law), are mutually independent. The coupling law admits a fixed–point reformulation: it is the unique bounded finitely additive solution of a Banach–contraction equation $f = T_\eta f$ determined by (μ, Π_R, η), with closed form $f_*(B) = \mu(B) + \frac{\eta}{1-\eta}\mu(\Pi_R^{-1}(B))$. Admissible structural models with fixed η form a category $\mathbf{Struct}_\eta$; under fiber measurability and either a finiteness (R–fin) or countability plus σ–additivity (R–ctbl) hypothesis, a quotient–factorization theorem reduces the classification problem to the identity–retraction case $\Pi_R = \mathrm{id}_X$, where each G–equivalence class C_k satisfies $\mu(C_k) \in \{0, (1-\eta)^{-1}\}$ under the hypotheses of Theorem 5.4.

Keywords: Axiomatic Systems[1]; Set Theory (ZFC); Measure Theory; Charges and Finitely Additive Measures; Idempotent Relations; Retraction Operators; Coupling Law; Banach Fixed-Point Theorem; Independence of Axioms; Categorical Structure; Quotient Factorization.

[1] This is an example

1 Introduction

We present a small ZFC–internal axiom system for abstract *pre-structural data*, tuples

$$\mathcal{D} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

consisting of: (i) a set algebra $(X, \mathcal{A})$ with a finitely additive measure $\mu : \mathcal{A} \to [0, \infty)$ and a finitely additive product charge $\mu^{\otimes 2}$ on $\mathcal{A} \otimes \mathcal{A}$ satisfying the rectangle rule; (ii) disjoint measurable sets $R, I \in \mathcal{A}$ and a map $\Pi_R : X \to R$ (not yet required to be idempotent); (iii) a measurable relation $G \subseteq X \times X$ (not yet required to be reflexive, symmetric, or idempotent); (iv) two scalars $E_0 > 0$ and $\eta \in [0, 1]$.

Three axioms tie these components together. Axiom I imposes projective structure (idempotence of Π_R and $\Pi_R|_R = \mathrm{id}_R$, plus measurability of preimages of measurable

subsets of R). Axiom II imposes closure structure on G (reflexivity, symmetry, and the idempotence $G \circ G = G$). Axiom III imposes quantitative coupling between measure, projection, and relation via three subclauses: conservation ($\mu(R) + \mu(I) = E_0 > 0$), projection–measure invariance ($\mu(\Pi_R^{-1}(B)) = \mu(B)$ for measurable $B \subseteq R$), and a coupling law on $(B \times X) \cap G$. A pre-structural datum satisfying Axioms I–III is called an *admissible structural model*.

Our goal is to isolate a minimal first–order framework in which projection, closure, and measure coexist and interact. *We do not assume $X = R \sqcup I$ at the axiomatic level;* only $R \cap I = \varnothing$ and measurability of $R \cup I$ are required. When the full partition $X = R \sqcup I$ is additionally imposed, the invariance clause of Axiom III forces $\mu(I) = 0$ (Proposition 4.3). Despite their minimality, the axioms yield explicit finite and countable admissible models as well as several structural consequences. We use finite additivity to keep the axiom system minimal; Section 7 records the additional constraints under σ–additivity.

On terminology. We adopt the name *curvature* for the idempotent symmetric relation G and for the derived two–point quantity $\mu^{\otimes 2}((B \times X) \cap G)$ as a convenient label for a bilinear, self–composing interaction pattern. This usage is analogical rather than differential–geometric: no Riemannian, sectional, or Ricci curvature is defined or used in this paper. The terminology emphasizes that $G \circ G = G$ imposes a closure–type rigidity on pair interactions, in the same way that curvature operators in geometry control how infinitesimal data compose. The reader for whom this analogy is unhelpful may replace "curvature" with "idempotent interaction" throughout without loss of content.

1.1 Motivation

Projection, closure, and measure are standard objects studied in different areas of mathematics (see e.g. [5, 6, 9] for measure theory, [7, 8] for charges, [10] for topology, [14] for projection/retraction operators in analysis, [13] for idempotent analysis, [16] for axiomatic foundations of measurement, and [17, 18, 20] for categorical viewpoints). The axioms place them into a single framework with direct interaction:

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G).$$

This coupling law does not appear in standard measure–theoretic or closure–theoretic axiomatizations. It couples the measure, projection, and closure components into a single axiom.

1.2 Relation to classical frameworks

Each component has a classical analogue:

- $G \circ G = G$ encodes an idempotent closure/equivalence–type behavior at the level of relations.
- $\Pi_R \circ \Pi_R = \Pi_R$ is the defining idempotence of a retraction.
- $(X, \mathcal{A}, \mu)$ is a finitely additive measure space.
- The invariance $\mu(\Pi_R^{-1}(B)) = \mu(B)$ for $B \subseteq R$ is analogous to a measure–preserving retraction onto R.

The key point is that these components coexist in one structure and are coupled by Axiom III.

1.3 Structural unification

The axioms provide a common interface for several classical viewpoints:

- **Measure/Probability**: if one normalizes $\mu(X) = 1$, the measure component can be read as a finitely additive probability.
- **Closure/Topology**: G induces a closure/connectedness pattern via its idempotence.
- **Equivalence**: on each G–connected component, idempotence forces transitivity along G–chains.
- **Projection**: Π_R preserves μ on measurable subsets of R (Axiom III).

Thus, measure, projection, and closure appear as compatible components of a single ZFC–internal structure.

1.4 Contributions

The contributions can be summarized as follows.

- A ZFC–internal axiom system (Section 3) and explicit model families establishing consistency (Section 5), including finite families with $\eta \neq 0$.
- Independence of the three axioms and of the three subclauses of Axiom III, by separating models (Theorem 6.7, Corollary 6.8).
- A fixed–point reformulation of the coupling law as the Banach–contraction equation $f = T_\eta f$, with closed form $f_*(B) = \mu(B) + \frac{\eta}{1-\eta}\mu(\Pi_R^{-1}(B))$ and a Neumann–series expansion (Theorem 4.8, Corollary 4.9).
- A quotient–factorization theorem reducing admissibility to the identity–retraction case $\Pi_R = \mathrm{id}_X$ under fiber measurability and either (R–fin) or (R–ctbl) (Theorem 8.18). In that case, under the hypotheses of Theorem 5.4, each G–equivalence class carries mass 0 or $(1-\eta)^{-1}$.

Section 7 records a side observation: under σ–additivity with $\mu(X) = 1$, the global instance of the coupling law forces $\eta = 0$; so nontrivial $\eta \neq 0$ families appear either in the finitely additive setting or for σ–additive measures not normalized as probabilities.

2 Language and basic objects

We recall standard notions used throughout the paper.

Definition 2.1 (Algebra of sets)**.** Let X be nonempty. A family $\mathcal{A} \subseteq \mathcal{P}(X)$ is an *algebra* if $\varnothing, X \in \mathcal{A}$ and $B_1, B_2 \in \mathcal{A}$ implies $B_1 \cup B_2 \in \mathcal{A}$ and $X \setminus B_1 \in \mathcal{A}$.

Definition 2.2 (Finitely additive measure)**.** A finitely additive measure is a map $\mu : \mathcal{A} \to [0, \infty)$ such that $\mu(\varnothing) = 0$ and $\mu(B_1 \sqcup B_2) = \mu(B_1) + \mu(B_2)$ whenever $B_1, B_2 \in \mathcal{A}$ are disjoint.

Definition 2.3 (Binary relations and composition)**.** A binary relation H on X is a subset of $X \times X$. Its composition with a relation K is

$$H \circ K = \{(x, z) : \exists y \ (x, y) \in H, \ (y, z) \in K\}.$$

The diagonal is $\Delta_X = \{(x, x) : x \in X\}$. A relation is *idempotent* if $H \circ H = H$.

Definition 2.4 (Retraction)**.** If $R \subseteq X$, a map $\Pi_R : X \to R$ is a *retraction* if $\Pi_R|_R = \mathrm{id}_R$. We also impose idempotence $\Pi_R \circ \Pi_R = \Pi_R$.

Definition 2.5 (Product algebra and product charge)**.** Given $(X, \mathcal{A}, \mu)$, the product algebra $\mathcal{A} \otimes \mathcal{A}$ is the algebra generated by rectangles $B_1 \times B_2$ with $B_i \in \mathcal{A}$.

A finitely additive set function $\mu^{\otimes 2} : \mathcal{A} \otimes \mathcal{A} \to [0, \infty)$ is called a *product charge* (for μ) if it satisfies the rectangle rule

$$\mu^{\otimes 2}(B_1 \times B_2) = \mu(B_1)\mu(B_2) \qquad (B_1, B_2 \in \mathcal{A}).$$

In the finitely additive setting, existence/uniqueness of such an extension is not automatic in full generality; whenever $\mu^{\otimes 2}$ is used below, it is treated as part of the model data satisfying the rectangle rule. (Under σ–additivity and σ–finiteness, the standard product charge exists uniquely; see Section 7.)

Lemma 2.6 (Atomic product charge exists)**.** *Assume X is finite or countable and $\mathcal{A} = \mathcal{P}(X)$. Suppose μ is atomic, i.e. there exists $m : X \to [0, \infty)$ such that $\mu(B) = \sum_{x \in B} m(x)$ for all $B \subseteq X$ (with the sum finite in the finite case). Define $\mu^{\otimes 2} : \mathcal{P}(X \times X) \to [0, \infty)$ by*

$$\mu^{\otimes 2}(S) := \sum_{(x,y) \in S} m(x)m(y) \qquad (S \subseteq X \times X).$$

Then $\mu^{\otimes 2}$ is finitely additive and satisfies the rectangle rule $\mu^{\otimes 2}(A \times B) = \mu(A)\mu(B)$ for all $A, B \subseteq X$. In particular, in the finite/countable atomic regimes used in Section 5, a product charge can be taken canonically.

Proof. Finite additivity follows from disjointness of sums. For rectangles,

$$\mu^{\otimes 2}(A \times B) = \sum_{x \in A} \sum_{y \in B} m(x)m(y) = \Big(\sum_{x \in A} m(x) \Big) \Big(\sum_{y \in B} m(y) \Big) = \mu(A)\mu(B).$$

□

3 The axioms

Definition 3.1 (Pre-structural datum)**.** A *pre-structural datum* is a tuple

$$\mathcal{D} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

such that:

- X is a nonempty set and $\mathcal{A} \subseteq \mathcal{P}(X)$ is an algebra;
- $\mu : \mathcal{A} \to [0, \infty)$ is a finitely additive measure;
- $\mu^{\otimes 2} : \mathcal{A} \otimes \mathcal{A} \to [0, \infty)$ is a finitely additive product charge satisfying the rectangle rule
$$\mu^{\otimes 2}(B_1 \times B_2) = \mu(B_1)\mu(B_2) \qquad (B_1, B_2 \in \mathcal{A});$$
- $R, I \in \mathcal{A}$ are disjoint and $R \cup I \in \mathcal{A}$;
- $\Pi_R : X \to R$ is a map;
- $G \subseteq X \times X$ lies in $\mathcal{A} \otimes \mathcal{A}$;
- $E_0 \in (0, \infty)$ and $\eta \in [0, 1]$ are scalars.

The pre-structural datum specifies only the raw signature of a structural model; the structural constraints (idempotence of Π_R, properties of G, compatibility with μ) are imposed separately by Axioms I–III below.

Axiom I (duality and projection). $R \cap I = \varnothing$ and $R \cup I \in \mathcal{A}$. Π_R is idempotent and $\Pi_R|_R = \mathrm{id}_R$. If $B \in \mathcal{A} \cap \mathcal{P}(R)$, then $\Pi_R^{-1}(B) \in \mathcal{A}$.

Axiom II (closed curvature). G is reflexive, symmetric, and idempotent: $G \circ G = G$.

Axiom III (Coupling). $\mu(R) + \mu(I) = E_0 > 0$. If $B \subseteq R$ is measurable then $\mu(\Pi_R^{-1}(B)) = \mu(B)$. For all $B \in \mathcal{A}$,

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G).$$

Remark 3.2. Since Π_R maps into R, one has $\Pi_R^{-1}(B) = \Pi_R^{-1}(B \cap R)$ for all $B \in \mathcal{A}$. Thus $\Pi_R^{-1}(B)$ is measurable whenever $B \cap R$ is, by Axiom I.

Definition 3.3 (Admissible structural model)**.** A pre-structural datum

$$\mathcal{D} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

is called an *admissible structural model* (or simply a *structural model*) if it satisfies Axioms I, II, and III. When we refer to $\mathcal{M}$ as a *structural model* without further qualification throughout the paper, admissibility is understood unless explicitly stated otherwise. The objects of the categories $\mathbf{Struct}_\eta$ and $\mathbf{Struct}_\eta^{\mathrm{id}}$ introduced in Section 8 are admissible structural models by convention.

4 Basic consequences

We record consequences of the axioms.

Proposition 4.1 (Elementary consequences of Axioms I and II)**.** *From Axioms I and II one has immediately:*

(i) (retraction fixed points) $\Pi_R(x) = x$ *if and only if* $x \in R$*;*

(ii) (iterated projection) $\Pi_R^n(x) = \Pi_R(x)$ *for every integer* $n \geq 1$*; in particular* $\Pi_R^n(x) \in R$*, so a point not initially in* I *is never sent into* I *by any iterate of* Π_R*;*

(iii) (G is an equivalence relation) *from reflexivity, symmetry, and idempotence* $G \circ G = G$*, transitivity follows: if* $(x, y), (y, z) \in G$ *then* $(x, z) \in G \circ G = G$*.*

Proof. (i) follows from $\Pi_R(X) \subseteq R$ and $\Pi_R|_R = \mathrm{id}_R$ (Axiom I). (ii) is a consequence of the idempotence $\Pi_R \circ \Pi_R = \Pi_R$: for $n = 1$ this is definitional, and for $n \geq 2$ we have $\Pi_R^n(x) = \Pi_R(\Pi_R^{n-1}(x)) = \Pi_R(x)$ by induction. (iii) is immediate from Axiom II as stated. □

Proposition 4.2 (Projection–measure invariance)**.** *If* $B \in \mathcal{A} \cap \mathcal{P}(R)$*, then* $\mu(\Pi_R^{-1}(B)) = \mu(B)$*.*

Proof. This is the second assertion of Axiom III. □

Proposition 4.3 (Support concentration under full partition)**.** *Assume in addition that* $X = R \sqcup I$*. Then Axiom III (projection–measure invariance) forces* $\mu(I) = 0$*.*

Proof. Take $B = R$ in $\mu(\Pi_R^{-1}(B)) = \mu(B)$ for measurable $B \subseteq R$. Since $\Pi_R^{-1}(R) = X$, we have $\mu(X) = \mu(R)$. By finite additivity on $X = R \sqcup I$, $\mu(X) = \mu(R) + \mu(I)$, hence $\mu(I) = 0$. □

Remark 4.4 (Observational weight and hidden sector)**.** Proposition 4.3 should be read as a design feature rather than a degeneracy: μ encodes *observational weight* carried by the representative sector R. The hidden sector I may be structurally present (e.g. as fibers of Π_R and through the interaction channel G), while being forced to carry no μ–mass under full-partition invariance. In particular, the axioms do *not* assume $X = R \sqcup I$; the full-partition hypothesis is an optional strengthening.

Example 4.5 (Why invariance forces zero mass on nontrivial fibers)**.** Let $X = \{r_0, r_1, a, b\}$, $R = \{r_0, r_1\}$, and $I = \{a, b\}$, and define a retraction $\Pi_R : X \to R$ by $\Pi_R|_R = \mathrm{id}_R$ together with $\Pi_R(a) = r_0$ and $\Pi_R(b) = r_1$. If $\mu(\{a\}) > 0$, then for $B = \{r_0\} \subseteq R$ one has $\Pi_R^{-1}(B) = \{r_0, a\}$ and hence

$$\mu(\Pi_R^{-1}(B)) = \mu(\{r_0\}) + \mu(\{a\}) > \mu(\{r_0\}) = \mu(B),$$

so the invariance $\mu(\Pi_R^{-1}(B)) = \mu(B)$ fails. Thus, under Axiom III, positive mass cannot be carried on nontrivial fibers of Π_R.

Corollary 4.6 (Baseline case $\eta = 0$)**.** *If $\eta = 0$, the coupling law of Axiom III reduces to*

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B) \qquad (\forall\, B \in \mathcal{A}),$$

so the observed curvature load coincides with the base measure.

Proof. Set $\eta = 0$ in the coupling law of Axiom III. □

4.1 Fixed-point characterization of the coupling law

We record an operator–theoretic reformulation of subclause (c) of Axiom III (the coupling law) as a fixed–point equation determining a unique bounded extension of μ under the projection Π_R.

Throughout this subsection we assume Axioms I and II hold, and that μ is bounded: $\mu(X) \le M < \infty$. Let

$$\mathcal{B} := \{\, f : \mathcal{A} \to \mathbb{R} \;\mid\; f \text{ finitely additive and bounded} \,\}$$

denote the space of bounded finitely additive signed charges, equipped with the supremum norm

$$\|f\|_\infty := \sup_{B \in \mathcal{A}} |f(B)|.$$

Under this norm $\mathcal{B}$ is a Banach space (see, e.g., [7, 9]). The subset $\mathcal{B}_+$ of nonnegative f is a closed convex cone in $\mathcal{B}$.

Definition 4.7 (Projection–update operator)**.** For $\eta \in [0, 1)$, define the operator $T_\eta : \mathcal{B} \to \mathcal{B}$ by

$$(T_\eta f)(B) := \mu(B) + \eta\, f(\Pi_R^{-1}(B)) \qquad (B \in \mathcal{A}).$$

That T_η is well–defined into $\mathcal{B}$ follows from three observations: (i) $\Pi_R^{-1}(B) \in \mathcal{A}$ for all $B \in \mathcal{A}$ (Remark 3.2); (ii) both μ and $f \circ \Pi_R^{-1}$ are finitely additive in B, hence so is $T_\eta f$; (iii) if $\|f\|_\infty < \infty$ then $|T_\eta f(B)| \le M + \eta\|f\|_\infty$, so $T_\eta f \in \mathcal{B}$. Moreover T_η preserves nonnegativity: if $f \ge 0$ on $\mathcal{A}$ and $\mu \ge 0$, then $T_\eta f \ge 0$, so $T_\eta(\mathcal{B}_+) \subseteq \mathcal{B}_+$.

Theorem 4.8 (Coupling law as unique fixed point)**.** *Assume Axioms I and II, and $\mu(X) \le M < \infty$. For $\eta \in [0, 1)$, the equation*

$$f = T_\eta f$$

has a unique solution $f_* \in \mathcal{B}$, given explicitly by

$$f_*(B) = \mu(B) + \frac{\eta}{1-\eta}\, \mu(\Pi_R^{-1}(B)) \qquad (B \in \mathcal{A}).$$

Moreover, $f_* \in \mathcal{B}_+$, and for any starting point $f_0 \in \mathcal{B}$ the iterates $f_n := T_\eta^n f_0$ satisfy

$$\|f_n - f_*\|_\infty \le \eta^n \|f_0 - f_*\|_\infty,$$

so $f_n \to f_*$ in norm.

Proof. Contraction. For $f_1, f_2 \in \mathcal{B}$ and any $B \in \mathcal{A}$,

$$|(T_\eta f_1)(B) - (T_\eta f_2)(B)| = \eta\, |f_1(\Pi_R^{-1}(B)) - f_2(\Pi_R^{-1}(B))| \le \eta \|f_1 - f_2\|_\infty.$$

Taking the supremum over B yields $\|T_\eta f_1 - T_\eta f_2\|_\infty \le \eta \|f_1 - f_2\|_\infty$. Since $\eta < 1$, T_η is a strict contraction on the Banach space $\mathcal{B}$. By Banach's fixed–point theorem there exists a unique $f_* \in \mathcal{B}$ with $f_* = T_\eta f_*$, together with the stated geometric convergence rate for iterates.

Explicit form. We verify that the stated closed form $f_*(B) = \mu(B) + \frac{\eta}{1-\eta}\mu(\Pi_R^{-1}(B))$ satisfies $f_* = T_\eta f_*$. The key identity is the *idempotence of the preimage map*:

$$\Pi_R^{-1}(\Pi_R^{-1}(B)) = \Pi_R^{-1}(B) \qquad (\forall B \in \mathcal{A}).$$

Indeed, for any $x \in X$, $x \in \Pi_R^{-1}(\Pi_R^{-1}(B))$ if and only if $\Pi_R(x) \in \Pi_R^{-1}(B)$, if and only if $\Pi_R(\Pi_R(x)) \in B$, if and only if $\Pi_R(x) \in B$ (by idempotence of Π_R from Axiom I, see Proposition 4.1(ii)), if and only if $x \in \Pi_R^{-1}(B)$. Hence

$$\begin{aligned}
(T_\eta f_*)(B) &= \mu(B) + \eta\, f_*(\Pi_R^{-1}(B)) \\
&= \mu(B) + \eta\Big[\mu(\Pi_R^{-1}(B)) + \tfrac{\eta}{1-\eta}\,\mu\big(\Pi_R^{-1}(\Pi_R^{-1}(B))\big)\Big] \\
&= \mu(B) + \eta\,\mu(\Pi_R^{-1}(B)) + \tfrac{\eta^2}{1-\eta}\,\mu(\Pi_R^{-1}(B)) \\
&= \mu(B) + \Big[\eta + \tfrac{\eta^2}{1-\eta}\Big]\mu(\Pi_R^{-1}(B)) \\
&= \mu(B) + \tfrac{\eta}{1-\eta}\,\mu(\Pi_R^{-1}(B)) = f_*(B).
\end{aligned}$$

Hence f_* is a fixed point; by uniqueness, it is the fixed point. Nonnegativity $f_* \ge 0$ is immediate since $\mu \ge 0$ and $\eta/(1-\eta) \ge 0$. □

Corollary 4.9 (Coupling law as identification of curvature load)**.** *Assume Axioms I and II, and $\mu(X) < \infty$. For $\eta \in [0,1)$, the coupling law (subclause (c) of Axiom III) is equivalent to the identification of the observed curvature load with the unique fixed point of T_η:*

$$\mu^{\otimes 2}\big((B \times X) \cap G\big) = f_*(B) = \mu(B) + \frac{\eta}{1-\eta}\,\mu(\Pi_R^{-1}(B)) \qquad (\forall B \in \mathcal{A}).$$

Proof. Let $\mathcal{C}(B) := \mu^{\otimes 2}((B \times X) \cap G)$. The coupling law asserts $\mathcal{C}(B) = \mu(B) + \eta\,\mathcal{C}(\Pi_R^{-1}(B))$ for all $B \in \mathcal{A}$, i.e. $\mathcal{C} = T_\eta \mathcal{C}$. Finite additivity of $\mathcal{C}$ in B follows from the rectangle rule and finite additivity of $\mu^{\otimes 2}$, applied to the disjoint decomposition $(B_1 \sqcup B_2) \times X = (B_1 \times X) \sqcup (B_2 \times X)$. Boundedness: $\mathcal{C}(B) \le \mu^{\otimes 2}(X \times X) = \mu(X)^2 < \infty$. Hence $\mathcal{C} \in \mathcal{B}$. By Theorem 4.8 the only fixed point of T_η in $\mathcal{B}$ is f_*, so $\mathcal{C} = f_*$. □

Remark 4.10 (Two compatibility conditions determining the coupling law)**.** Corollary 4.9 expresses the coupling law as the unique bounded finitely additive solution of the operator equation $f = T_\eta f$. The operator T_η encodes two conditions on a quantity f defined on $\mathcal{A}$:

- *Baseline agreement with μ*: for $\eta = 0$ one has $T_0 f = \mu$ and $f_* = \mu$.
- *Self–similar propagation along Π_R*: the correction to μ propagates under the preimage map Π_R^{-1}, scaled by η.

Requiring the curvature load $\mathcal{C}(B) = \mu^{\otimes 2}((B \times X) \cap G)$ to satisfy both conditions determines the coupling law uniquely once $(\mathcal{A}, \mu, \Pi_R, \eta)$ are fixed.

Remark 4.11 (Neumann–series expansion)**.** Formally iterating T_η from any $f_0 \in \mathcal{B}$ and using Banach convergence, one obtains the Neumann–series expression

$$f_*(B) = \sum_{n=0}^{\infty} \eta^n\, \mu(\Pi_R^{-n}(B)),$$

convergent in $\|\cdot\|_\infty$ for $\eta \in [0,1)$. Using the preimage idempotence $\Pi_R^{-n}(B) = \Pi_R^{-1}(B)$ for $n \geq 1$ (derived in the proof of Theorem 4.8), the series telescopes to

$$f_*(B) = \mu(B) + \mu(\Pi_R^{-1}(B)) \sum_{n=1}^{\infty} \eta^n = \mu(B) + \frac{\eta}{1-\eta}\,\mu(\Pi_R^{-1}(B)),$$

recovering the closed form. The term $\mu(B)$ is the baseline ($n = 0$) contribution; the geometric sum $\frac{\eta}{1-\eta}$ accumulates all iterated projection corrections.

Remark 4.12 (Relation to the classification when $\Pi_R = \mathrm{id}$)**.** When $\Pi_R = \mathrm{id}_X$, one has $\Pi_R^{-1}(B) = B$, so the fixed–point formula collapses to

$$f_*(B) = \frac{\mu(B)}{1-\eta} \qquad (B \text{ measurable}).$$

In regimes where the full hypotheses of Theorem 5.4 hold for the relevant identity–retraction core, this yields the blockwise normalization law: every positive–mass G–equivalence class C_k satisfies $\mu(C_k) = (1-\eta)^{-1}$.

4.2 Observable reduction under global decoupling

Definition 4.13 (Observed curvature load)**.** For measurable $B \subseteq R$, define

$$\mathcal{C}_R(B) := \mu^{\otimes 2}\big((B \times X) \cap G\big).$$

Theorem 4.14 (Exact reduction on R under global decoupling)**.** *Let $\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ be a structural model with $\eta \in [0,1)$. Assume the global decoupling condition*

$$\mu^{\otimes 2}\big(((\Pi_R^{-1}(B) \setminus B) \times X) \cap G\big) = 0 \qquad (\forall\, B \subseteq R \text{ measurable}).$$

Then for every measurable $B \subseteq R$:

*(a) (*support reduction*) $\mu^{\otimes 2}\big(G \setminus (R \times R)\big) = 0$ and*

$$\mu^{\otimes 2}\big((B \times X) \cap G\big) = \mu^{\otimes 2}\big((B \times R) \cap (G \cap (R \times R))\big).$$

*(b) (*closed form on R*) the coupling law closes on R:*

$$\mu^{\otimes 2}\big((B \times X) \cap G\big) = \frac{\mu(B)}{1-\eta}.$$

Proof. Fix measurable $B \subseteq R$.

(*Support reduction.*) Apply the decoupling condition with $B = R$. Since $\Pi_R^{-1}(R) = X$, we have $\Pi_R^{-1}(R) \setminus R = X \setminus R$, hence

$$0 = \mu^{\otimes 2}\big(((X \setminus R) \times X) \cap G\big).$$

By symmetry of G, also $\mu^{\otimes 2}((X \times (X \setminus R)) \cap G) = 0$, and therefore $\mu^{\otimes 2}(G \setminus (R \times R)) = 0$. In particular, $(B \times X) \cap G$ and $(B \times R) \cap (G \cap (R \times R))$ differ by a $\mu^{\otimes 2}$–null set, yielding the stated equality.

(*Closed form.*) Since $B \subseteq \Pi_R^{-1}(B)$ and $\Pi_R^{-1}(B) = B \sqcup (\Pi_R^{-1}(B) \setminus B)$, we have a disjoint union

$$(\Pi_R^{-1}(B) \times X) \cap G = ((B \times X) \cap G) \;\sqcup\; \big(((\Pi_R^{-1}(B) \setminus B) \times X) \cap G\big).$$

By the decoupling hypothesis, the second term has $\mu^{\otimes 2}$–measure zero, hence

$$\mu^{\otimes 2}\big((\Pi_R^{-1}(B) \times X) \cap G\big) = \mu^{\otimes 2}\big((B \times X) \cap G\big).$$

Insert this into the coupling law (Axiom III) for $B \subseteq R$:

$$\mu^{\otimes 2}\big((B \times X) \cap G\big) = \mu(B) + \eta\, \mu^{\otimes 2}\big((\Pi_R^{-1}(B) \times X) \cap G\big),$$

to obtain $(1-\eta)\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)$, i.e. the stated closed form. □

5 Models and consistency

We construct explicit models.

5.1 Finite models

Theorem 5.1. *There exists a finite structural model. In particular, the axiom set is satisfiable in ZFC.*

Proof. Let X be finite and nonempty, and choose disjoint nonempty subsets $R, I \subseteq X$ Let X be finite and nonempty, and choose disjoint nonempty subsets $R, I \subseteq X$ with $R \cup I = X$. Let $\mathcal{A} = \mathcal{P}(X)$. Fix $r_0 \in R$ and define $\Pi_R : X \to R$ by

$$\Pi_R(x) = \begin{cases} x, & x \in R, \\ r_0, & x \in I. \end{cases}$$

Then Π_R is idempotent and $\Pi_R|_R = \mathrm{id}_R$. Since $\mathcal{A}$ is the full power set, $\Pi_R^{-1}(B) \in \mathcal{A}$ for all $B \subseteq R$, so Axiom I holds.

Define $G = \Delta_X \subseteq X \times X$. Then G is reflexive, symmetric, and idempotent; also $G \in \mathcal{A} \otimes \mathcal{A}$. Thus Axiom II holds.

Choose weights $m(x) \in \{0, 1\}$ such that $m(x) = 0$ for all $x \in I$ and $\sum_{x \in R} m(x) > 0$. Define $\mu : \mathcal{A} \to [0, \infty)$ by

$$\mu(B) = \sum_{x \in B} m(x).$$

Then μ is finitely additive, $\mu(I) = 0$, and $\mu(R) > 0$. Set

$$E_0 := \mu(X) = \mu(R) \in (0, \infty).$$

This gives the conservation identity $\mu(R) + \mu(I) = E_0 > 0$.

If $B \subseteq R$, then

$$\Pi_R^{-1}(B) = \begin{cases} B, & r_0 \notin B, \\ B \cup I, & r_0 \in B, \end{cases}$$

so

$$\mu(\Pi_R^{-1}(B)) = \begin{cases} \mu(B), & r_0 \notin B, \\ \mu(B) + \mu(I), & r_0 \in B, \end{cases} = \mu(B),$$

because $\mu(I) = 0$. Thus the projection–measure invariance part of Axiom III holds.

Using the product charge $\mu^{\otimes 2}$, For any $B \subseteq X$,

$$(B \times X) \cap G = \{(x, x) : x \in B\},$$

and therefore

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{x \in B} \mu(\{x\})^2 = \sum_{x \in B} m(x)^2 = \sum_{x \in B} m(x) = \mu(B),$$

using $m(x)^2 = m(x)$ for $m(x) \in \{0, 1\}$. Taking $\eta = 0$, the coupling law in Axiom III reduces to $\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)$, which we have just verified. Thus we obtain a finite structural model satisfying all axioms. □

5.2 Further finite models with $\eta \neq 0$

The preceding constructions set $\eta = 0$ for simplicity. We now record explicit finite models and families in which $\eta \neq 0$.

Theorem 5.2 (Finite model with $\eta \neq 0$)**.** *Let $X = \{r_0, r_1, i\}$ and define a measurable partition $X = R \sqcup I$ by*

$$R = \{r_0, r_1\}, \qquad I = \{i\}.$$

Define a projection (retraction) $\Pi_R : X \to R$ by

$$\Pi_R(x) = \begin{cases} x, & x \in R, \\ r_0, & x = i \in I. \end{cases}$$

Let $G = \Delta_X = \{(x, x) : x \in X\} \subseteq X \times X$. Fix any $\eta \in [0, 1)$ with $\eta \neq 0$. Define a finitely additive measure μ on $\mathcal{P}(X)$ by point masses

$$m(i) = 0, \qquad m(r_0) = m(r_1) = \frac{1}{1-\eta}, \qquad \mu(B) := \sum_{x \in B} m(x).$$

Define the product charge $\mu^{\otimes 2}$ on $\mathcal{P}(X \times X)$ by rectangle masses $\mu^{\otimes 2}(A \times B) := \mu(A)\mu(B)$ and finite additivity.

Then $\mathcal{M} = (X, \mathcal{P}(X), \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ satisfies Axioms I–III, with $E_0 = \mu(R) + \mu(I) = \frac{2}{1-\eta} > 0$.

Proof. Axiom I and Axiom II are immediate from the definitions. For Axiom III, conservation holds by the choice of E_0. If $B \subseteq R$, then $\Pi_R^{-1}(B)$ equals B or $B \cup \{i\}$ depending on whether $r_0 \in B$, and $m(i) = 0$ yields $\mu(\Pi_R^{-1}(B)) = \mu(B)$.

For the coupling law, note that $(B \times X) \cap \Delta_X = \{(x, x) : x \in B\}$, so

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{x \in B} m(x)^2.$$

Similarly, $\mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G) = \sum_{x \in \Pi_R^{-1}(B)} m(x)^2$, and the extra term (if any) is $m(i)^2 = 0$. Thus the coupling law reduces to checking for each $r \in R$:

$$m(r)^2 = m(r) + \eta\, m(r)^2 \iff (1-\eta) m(r)^2 = m(r),$$

which holds for $m(r) = \frac{1}{1-\eta}$. □

Proposition 5.3 (A finite model with $\eta \neq 0$ and non-diagonal curvature)**.** *Let $X = \{r_0, r_1\}$ and take $R = X$, $I = \varnothing$, $\Pi_R = \mathrm{id}_X$. Fix any $\eta \in [0, 1)$, $\eta \neq 0$. Define μ by*

$$m(r_0) = m(r_1) = \frac{1}{2(1-\eta)}, \qquad \mu(B) = \sum_{x \in B} m(x),$$

so that $\mu(X) = \frac{1}{1-\eta}$. Let $G = X \times X$ (the total relation on X). Let $\mu^{\otimes 2}$ be the canonical atomic product charge on $\mathcal{P}(X \times X)$ induced by μ (Lemma 2.6). Then $\mathcal{M} = (X, \mathcal{P}(X), \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ satisfies Axioms I–III with $E_0 = \mu(R) = \frac{1}{1-\eta}$, and G is non-diagonal.

Proof. Invariance is immediate since $\Pi_R = \mathrm{id}_X$. For any $B \subseteq X$, $(B \times X) \cap G = B \times X$, hence

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)\mu(X).$$

Also $\Pi_R^{-1}(B) = B$, so the coupling law becomes $\mu(B)\mu(X) = \mu(B) + \eta\, \mu(B)\mu(X)$. If $\mu(B) > 0$ we cancel $\mu(B)$ to obtain $(1-\eta)\mu(X) = 1$, which holds by construction; if $\mu(B) = 0$ the identity is trivial. □

Theorem 5.4 (Canonical classification when $\Pi_R = \mathrm{id}$)**.** *Assume $R = X$, $I = \varnothing$, and $\Pi_R = \mathrm{id}_X$. Let $\eta \in [0,1)$. Let $G \subseteq X \times X$ be reflexive, symmetric, and idempotent ($G \circ G = G$). Then G is an equivalence relation on X; denote its equivalence classes by $(C_k)_{k\in K}$, so that*

$$G = \bigcup_{k\in K} (C_k \times C_k) \qquad \textit{(disjoint union of blocks).}$$

Let μ be a finitely additive measure on $\mathcal{P}(X)$ and let $\mu^{\otimes 2}$ be the product charge on the product algebra $\mathcal{P}(X) \otimes \mathcal{P}(X)$, satisfying the rectangle rule of Definition 2.5. Suppose further that one of the following summability hypotheses holds, so that the block formula below is well–defined:

(K–fin) *the index set K is finite; or*

(K–ctbl) *K is countable, μ is σ–additive (so $\mathcal{P}(X)$ is treated as a σ–algebra) and σ–finite on X, and $\mu^{\otimes 2}$ is the standard σ–additive product measure on the product σ–algebra $\mathcal{P}(X) \otimes \mathcal{P}(X)$ (which then exists and is uniquely characterized by the rectangle rule).*

Then the coupling law

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B) + \eta\, \mu^{\otimes 2}((B \times X) \cap G) \qquad (\forall\, B \subseteq X \ \textit{measurable})$$

holds if and only if, for every class C_k with $\mu(C_k) > 0$,

$$\mu(C_k) = \frac{1}{1-\eta}.$$

Equivalently, for all measurable $B \subseteq X$ one has the explicit block formula

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{k\in K} \mu(B \cap C_k)\, \mu(C_k),$$

and the coupling law is precisely the blockwise normalization law above.

Proof. First, G is transitive: if $(x,y) \in G$ and $(y,z) \in G$, then $(x,z) \in G \circ G = G$. With reflexivity and symmetry, G is an equivalence relation, hence decomposes into blocks (C_k) and $G = \bigcup_k (C_k \times C_k)$.

For any measurable $B \subseteq X$,

$$(B \times X) \cap G = \bigsqcup_{k\in K} \big((B \cap C_k) \times C_k\big),$$

a disjoint union. Under (K–fin) finite additivity of $\mu^{\otimes 2}$ together with the rectangle rule gives

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{k\in K} \mu(B \cap C_k)\, \mu(C_k)$$

immediately. Under (K–ctbl) the same identity follows from σ–additivity of $\mu^{\otimes 2}$ applied to the countable disjoint union together with the rectangle rule. In either case we obtain the stated block formula.

Since $\Pi_R = \mathrm{id}_X$, the coupling law reduces to

$$(1-\eta)\, \mu^{\otimes 2}((B \times X) \cap G) = \mu(B) \qquad (\forall\, B \subseteq X \text{ measurable}).$$

Taking $B = C_j$ yields $(1-\eta)\mu(C_j)^2 = \mu(C_j)$, hence $\mu(C_j) = 0$ or $\mu(C_j) = \frac{1}{1-\eta}$. Thus all positive-mass classes satisfy the normalization $\mu(C_k) = \frac{1}{1-\eta}$. Conversely, if this holds for all positive-mass classes, then the block formula implies $\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)/(1-\eta)$ for all measurable B, so the coupling law holds. $\square$

Remark 5.5 (On the summability hypothesis)**.** The block formula of Theorem 5.4 involves a sum over K, which can fail to make sense in the purely finitely additive setting when K is infinite: arbitrary countable disjoint unions need not be countably additive for a finitely additive $\mu^{\otimes 2}$. The hypotheses (K–fin) or (K–ctbl) are exactly what guarantees the block formula as an identity; under neither hypothesis, the theorem holds only in the weaker form "for each finite sub-collection of classes, the corresponding partial block formula holds", which is insufficient for the global coupling–law characterization. We therefore always state downstream results (including Corollary 8.19 and Theorem 8.18) under these summability hypotheses when the block formula is invoked.

Corollary 5.6 (Total mass in the fully positive block case)**.** *Under the hypotheses of Theorem 5.4, suppose that X has finitely many G–equivalence classes $C_1, \dots, C_m$ and that $\mu(C_j) > 0$ for all j. Then*

$$\mu(X) = \frac{m}{1-\eta} \qquad \text{and} \qquad \mu^{\otimes 2}(G) = \sum_{j=1}^{m} \mu(C_j)^2 = \frac{m}{(1-\eta)^2}.$$

Moreover, for every $B \subseteq X$ one has the closed form

$$\mu^{\otimes 2}((B \times X) \cap G) = \frac{\mu(B)}{1-\eta}.$$

Proof. By Theorem 5.4, $\mu(C_j) = \frac{1}{1-\eta}$ for each j. Since $X = \bigsqcup_{j=1}^{m} C_j$, finite additivity gives $\mu(X) = \sum_{j=1}^{m} \mu(C_j) = \frac{m}{1-\eta}$. Also $G = \bigcup_{j=1}^{m} (C_j \times C_j)$ is a disjoint union, hence

$$\mu^{\otimes 2}(G) = \sum_{j=1}^{m} \mu(C_j)\mu(C_j) = \sum_{j=1}^{m} \mu(C_j)^2 = \frac{m}{(1-\eta)^2}.$$

Finally, the same theorem yields $\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)/(1-\eta)$ for all $B \subseteq X$. □

Remark 5.7. Theorem 5.4 subsumes the block–curvature family (Theorem 5.8) by deriving the block decomposition directly from the axioms on G.

Theorem 5.8 (Block-curvature family for $\eta \neq 0$)**.** *Let X be finite and let $\{C_1, \dots, C_m\}$ be a partition of X into nonempty blocks. Set $R = X$, $I = \varnothing$, $\Pi_R = \mathrm{id}_X$, and define*

$$G := \bigcup_{k=1}^{m} (C_k \times C_k) \subseteq X \times X.$$

Fix $\eta \in [0, 1)$ with $\eta \neq 0$. Let μ be any finitely additive measure on $\mathcal{P}(X)$ and define $\mu^{\otimes 2}$ on rectangles by $\mu^{\otimes 2}(A \times B) = \mu(A)\mu(B)$.

Then the coupling law holds for all $B \subseteq X$ if and only if

$$\mu(C_k) = \frac{1}{1-\eta} \qquad \text{for all } k = 1, \dots, m.$$

In particular, at fixed $\eta \neq 0$ the coupling law imposes exactly m scalar constraints (one per G-component), and imposes no further constraint on the distribution of μ inside each block.

Proof. For any $B \subseteq X$,

$$(B \times X) \cap G = \bigcup_{k=1}^{m} \big((B \cap C_k) \times C_k\big)$$

is a disjoint union, hence

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{k=1}^{m} \mu(B \cap C_k)\, \mu(C_k).$$

Since $\Pi_R^{-1}(B) = B$, the coupling law is equivalent to

$$(1-\eta) \sum_{k=1}^{m} \mu(B \cap C_k)\, \mu(C_k) = \mu(B) \qquad (\forall B \subseteq X).$$

Taking $B = C_j$ yields $(1-\eta)\mu(C_j)^2 = \mu(C_j)$, so $\mu(C_j) = 0$ or $\mu(C_j) = \frac{1}{1-\eta}$. For nontrivial blocks we take the positive solution and obtain necessity. Conversely, if $\mu(C_k) = \frac{1}{1-\eta}$ for all k, then

$$\mu^{\otimes 2}((B \times X) \cap G) = \frac{1}{1-\eta} \sum_{k=1}^{m} \mu(B \cap C_k) = \frac{1}{1-\eta} \mu(B),$$

which satisfies the coupling law identically. □

Remark 5.9 (Interpretation: blockwise normalization law)**.** For $\Pi_R = \mathrm{id}$, the coupling law acts as a *blockwise normalization law*: it fixes the total mass of each G–connected component to $(1-\eta)^{-1}$, while leaving internal distributions free. The endpoint $\eta = 1$ would force $\mu(C_k) = +\infty$ and is therefore excluded in these finite families.

We next give a finite model with nontrivial curvature.

Proposition 5.10 (Finite class model with nontrivial curvature)**.** *Let X be a finite nonempty set, partitioned into nonempty equivalence classes $(C_j)_{j \in J}$. For each j, choose a distinguished representative $r_j \in C_j$ and set*

$$R := \{r_j : j \in J\}, \qquad I := X \setminus R.$$

Let $\mathcal{A} = \mathcal{P}(X)$. Choose weights $w_j \in \{0,1\}$ with $\sum_j w_j > 0$ and define a finitely additive measure $\mu : \mathcal{A} \to [0,\infty)$ by

$$\mu(B) := \sum_{j \in J : r_j \in B} w_j.$$

Define $\Pi_R : X \to R$ by $\Pi_R(x) = r_j$ whenever $x \in C_j$, and define

$$G := \bigcup_{j \in J} (C_j \times C_j) \subseteq X \times X.$$

Set $E_0 := \mu(X)$ and $\eta := 0$. Let $\mu^{\otimes 2}$ be the canonical atomic product charge on $\mathcal{P}(X \times X)$ induced by μ (Lemma 2.6). Then

$$\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

is a structural model. In particular, there exist finite models with nontrivial curvature.

Proof. By construction, $R, I \in \mathcal{A}$, $R \cap I = \varnothing$, and $R \cup I = X$. The map Π_R restricts to the identity on R, and $\Pi_R \circ \Pi_R = \Pi_R$ because every x is sent to its class representative. Preimages of subsets of R are unions of classes, hence measurable. Thus Axiom I holds.

The relation G is reflexive, symmetric, and idempotent since it is the union of equivalence relations on the blocks C_j. Clearly $G \in \mathcal{A} \otimes \mathcal{A}$, so Axiom II holds.

For Axiom III, note first that

$$\mu(R) + \mu(I) = \mu(R) = \sum_j w_j =: E_0 > 0.$$

If $B \subseteq R$, then

$$\Pi_R^{-1}(B) = \bigcup_{j:r_j \in B} C_j,$$

and therefore

$$\mu(\Pi_R^{-1}(B)) = \sum_{j:r_j \in B} \mu(C_j) = \sum_{j:r_j \in B} w_j = \mu(B).$$

Thus the projection–measure invariance holds.

Equip $X \times X$ with the product charge $\mu^{\otimes 2}$. For any $B \subseteq X$, write $B_j := B \cap C_j$. Then

$$(B \times X) \cap G = \bigcup_{j \in J} (B_j \times C_j),$$

disjoint union. Hence

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{j \in J} \mu(B_j)\, \mu(C_j) = \sum_{j:r_j \in B} w_j^2 = \sum_{j:r_j \in B} w_j = \mu(B),$$

using $\mu(B_j) = w_j$ if $r_j \in B_j$ and 0 otherwise, and $w_j^2 = w_j$ for $w_j \in \{0, 1\}$. With $\eta = 0$, the coupling law reduces to $\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)$. Thus $\mathcal{M}$ satisfies all axioms and has nontrivial curvature whenever some class C_j has cardinality > 1. □

5.3 Countable model

Proposition 5.11. *There exists a structural model with X countably infinite.*

Proof. Let $\mathcal{M}_0 = (X_0, \mathcal{A}_0, \mu_0, \mu_0^{\otimes 2}, R_0, I_0, \Pi_{R,0}, G_0, E_{0,0}, \eta_0)$ be a finite structural model as in Theorem 5.1, with $\eta_0 = 0$. Choose a countably infinite set Y disjoint from X_0 and set

$$X := X_0 \sqcup Y.$$

Define $\mathcal{A}$ to be the algebra generated by $\mathcal{A}_0$ together with all subsets of Y. More concretely, $\mathcal{A}$ consists of all sets of the form

$$B_0 \sqcup B_Y, \qquad B_0 \in \mathcal{A}_0,\ B_Y \subseteq Y.$$

Define $\mu : \mathcal{A} \to [0, \infty)$ by

$$\mu(B_0 \sqcup B_Y) := \mu_0(B_0),$$

so that all points of Y have measure zero. Set

$$R := R_0, \qquad I := I_0 \sqcup Y.$$

Define $\Pi_R : X \to R$ by

$$\Pi_R(x) := \begin{cases} \Pi_{R,0}(x), & x \in X_0, \\ r_0, & x \in Y, \end{cases}$$

where $r_0 \in R_0$ is the distinguished point used in the finite model. Define $G \subseteq X \times X$ by

$$G := G_0 \cup \Delta_Y,$$

where $\Delta_Y = \{(y,y) : y \in Y\}$. Set $E_0 := \mu(R) + \mu(I) = \mu_0(R_0) + \mu_0(I_0) = E_{0,0}$ and $\eta := 0$.

We verify the axioms. Axiom I holds since $R, I \in \mathcal{A}$, $R \cap I = \varnothing$, $R \cup I = X$, Π_R is idempotent and equals $\Pi_{R,0}$ on R_0. Measurability of preimages for subsets of R follows from the definition of $\mathcal{A}$.

Axiom II holds because G_0 and Δ_Y are both reflexive, symmetric, and idempotent, and their union is block–diagonal with respect to the decomposition $X = X_0 \sqcup Y$. Moreover, $G \in \mathcal{A} \otimes \mathcal{A}$.

For Axiom III, note first that

$$\mu(R) + \mu(I) = \mu_0(R_0) + \mu_0(I_0) = E_{0,0} > 0.$$

If $B \subseteq R$ is measurable, then $B \subseteq R_0$ and

$$\Pi_R^{-1}(B) = \Pi_{R,0}^{-1}(B) \sqcup (Y \text{ if } r_0 \in B \text{ else } \varnothing),$$

so

$$\mu(\Pi_R^{-1}(B)) = \mu_0(\Pi_{R,0}^{-1}(B)) = \mu_0(B) = \mu(B),$$

using the finite model's Axiom III. Thus the projection–measure invariance holds.

Equip $X \times X$ with the product charge $\mu^{\otimes 2}$. If $B \in \mathcal{A}$ and we write $B = B_0 \sqcup B_Y$ with $B_0 \in \mathcal{A}_0$, $B_Y \subseteq Y$, then

$$(B \times X) \cap G = \big((B_0 \times X_0) \cap G_0\big) \ \sqcup \ \{(y,y) : y \in B_Y\},$$

and

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu_0^{\otimes 2}((B_0 \times X_0) \cap G_0) + \sum_{y \in B_Y} \mu(\{y\})^2 = \mu_0^{\otimes 2}((B_0 \times X_0) \cap G_0),$$

since $\mu(\{y\}) = 0$ for $y \in Y$. Similarly,

$$\mu(B) = \mu_0(B_0).$$

With $\eta = 0$, Axiom III reduces to

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B),$$

which we have just verified using the corresponding identity in the finite model. Thus $\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ is a countably infinite structural model. □

6 Independence of the axioms

In this section we establish that the three axioms of Section 3 are mutually independent: no axiom follows from the remaining two. Together with the model constructions of Section 5, which establish consistency, this shows that the axiom system is both consistent and minimal, justifying the word "minimal" used in the abstract and Section 1.4.

Independence is established in the model-theoretic sense: for each axiom $\mathsf{X} \in \{\mathrm{I}, \mathrm{II}, \mathrm{III}\}$ we exhibit a structure over the same signature which satisfies the other two axioms but fails X. We also record the stronger result that the three subclauses of Axiom III (conservation, projection–measure invariance, coupling law) are themselves mutually independent within Axiom III.

6.1 Separating models for Axioms I, II, III

The constructions below follow a minimal relaxation principle: each separating datum $\mathcal{D}_{\neg\mathsf{X}}$ is obtained from a baseline finite model (as in Section 5) by modifying exactly the feature asserted by the axiom under analysis, leaving the remaining structural features intact.

Proposition 6.1 (Independence of Axiom I)**.** *There exists a pre-structural datum $\mathcal{D}_{\neg\mathrm{I}}$ over the ambient signature of Definition 3.1 which satisfies Axioms II and III but fails Axiom I.*

Proof. Let $X = \{a, b, c\}$ and $\mathcal{A} = \mathcal{P}(X)$. Set

$$R := \{a, b\}, \qquad I := \{c\},$$

so $R \cap I = \varnothing$ and $R \cup I = X \in \mathcal{A}$. Define the map $\Pi_R : X \to R$ by

$$\Pi_R(a) := b, \qquad \Pi_R(b) := a, \qquad \Pi_R(c) := a.$$

Then $\Pi_R(X) \subseteq R$, but $\Pi_R|_R \neq \mathrm{id}_R$ (it swaps a and b) and $\Pi_R \circ \Pi_R \neq \Pi_R$ (since $\Pi_R(\Pi_R(a)) = \Pi_R(b) = a \neq b = \Pi_R(a)$). Hence Axiom I fails.

Set $G := \Delta_X$, which is reflexive, symmetric, and idempotent on any set; hence Axiom II holds.

Define $\mu : \mathcal{A} \to [0, \infty)$ by the point weights $m(a) = m(b) := 1$ and $m(c) := 0$, and let $\mu^{\otimes 2}$ be the canonical atomic product charge from Lemma 2.6. Set $\eta := 0$ and $E_0 := \mu(R) + \mu(I) = 2 > 0$, so Axiom III(a) (conservation) holds.

For projection–measure invariance, take any $B \subseteq R$ measurable:

$$\Pi_R^{-1}(\{a\}) = \{b, c\}, \quad \Pi_R^{-1}(\{b\}) = \{a\}, \quad \Pi_R^{-1}(R) = X,$$

and direct computation yields $\mu(\Pi_R^{-1}(B)) = \mu(B)$ for each $B \subseteq R$; in particular

$$\mu(\Pi_R^{-1}(\{a\})) = \mu(\{b, c\}) = 1 + 0 = 1 = \mu(\{a\}).$$

Hence Axiom III(b) holds.

For the coupling law with $\eta = 0$, since $G = \Delta_X$, for any $B \subseteq X$

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{x \in B} m(x)^2 = \sum_{x \in B} m(x) = \mu(B),$$

using $m(x) \in \{0, 1\}$ so $m(x)^2 = m(x)$. Hence Axiom III(c) holds. All of Axiom III is therefore satisfied, while Axiom I fails. □

Proposition 6.2 (Independence of Axiom II)**.** *There exists a pre-structural datum $\mathcal{D}_{\neg\mathrm{II}}$ over the ambient signature of Definition 3.1 which satisfies Axioms I and III but fails Axiom II.*

Proof. Let $X = \{a, b, c\}$ and $\mathcal{A} = \mathcal{P}(X)$. Set $R := X$, $I := \varnothing$, and $\Pi_R := \mathrm{id}_X$. Then Π_R is idempotent, $\Pi_R|_R = \mathrm{id}_R$, and preimages of measurable subsets are measurable. Hence Axiom I holds.

Define

$$G := \{(a, a), (b, b), (c, c), (a, b), (b, a), (b, c), (c, b)\} \subseteq X \times X.$$

Then G is reflexive ($\Delta_X \subseteq G$) and symmetric. However, G is not idempotent: the pair (a, c) is obtained via

$$(a, b) \in G, \quad (b, c) \in G \implies (a, c) \in G \circ G,$$

but $(a, c) \notin G$, so $G \circ G \supsetneq G$. Hence Axiom II fails.

Define μ by $m(a) := 1$, $m(b) := 0$, $m(c) := 1$, and let $\mu^{\otimes 2}$ be the atomic product charge. Set $\eta := 0$ and $E_0 := \mu(R) + \mu(I) = 2 > 0$, so Axiom III(a) holds. Since $\Pi_R = \mathrm{id}_X$, projection–measure invariance (Axiom III(b)) is trivial.

For the coupling law, we compute $\mu^{\otimes 2}((B \times X) \cap G)$ for each singleton B. The nonzero contributions come only from pairs $(x, y) \in G$ with $m(x)m(y) \neq 0$, i.e. with $x, y \in \{a, c\}$. The only such pairs in G are (a, a) and (c, c) (since $(a, b), (b, c)$ etc. involve b). Hence for each $B \subseteq X$:

$$\begin{aligned} B = \{a\} :\ & \mu^{\otimes 2}((B \times X) \cap G) = m(a)^2 = 1 = \mu(B), \\ B = \{b\} :\ & \mu^{\otimes 2}((B \times X) \cap G) = 0 = \mu(B), \\ B = \{c\} :\ & \mu^{\otimes 2}((B \times X) \cap G) = m(c)^2 = 1 = \mu(B), \end{aligned}$$

and finite additivity extends this to all $B \subseteq X$. With $\eta = 0$ and $\Pi_R = \mathrm{id}_X$, the coupling law reduces to $\mu^{\otimes 2}((B \times X) \cap G) = \mu(B)$, which we have verified. Hence Axiom III(c) holds. All of Axiom III is therefore satisfied, while Axiom II fails. □

Remark 6.3 (On the $\mu^{\otimes 2}$-invisibility of the failure of Axiom II)**.** In the datum $\mathcal{D}_{\neg\mathrm{II}}$ above, the failure of $G \circ G = G$ occurs at the pair (a, c), which carries positive $\mu^{\otimes 2}$-weight $(m(a)m(c) = 1)$ but lies outside G. The separating witness is set-theoretic ($G \circ G \supsetneq G$ as subsets of $X \times X$), not measure-theoretic. This reflects the fact that Axiom II is a structural rather than measure-theoretic statement; the failure is visible at the level of the relation G itself, regardless of whether the measure detects it through the coupling law.

Proposition 6.4 (Independence of Axiom III)**.** *There exists a pre-structural datum $\mathcal{D}_{\neg\mathrm{III}}$ over the ambient signature of Definition 3.1 which satisfies Axioms I and II but fails Axiom III.*

Proof. Let $X = \{r, i\}$ and $\mathcal{A} = \mathcal{P}(X)$. Set $R := \{r\}$, $I := \{i\}$, and define $\Pi_R : X \to R$ by $\Pi_R(r) := r$, $\Pi_R(i) := r$. Then Π_R is idempotent, $\Pi_R|_R = \mathrm{id}_R$, and preimages of measurable subsets of R are measurable. Hence Axiom I holds.

Set $G := \Delta_X$, satisfying Axiom II.

Define μ by $m(r) := m(i) := 1$, and let $\mu^{\otimes 2}$ be the atomic product charge. Set $\eta := 0$ and $E_0 := \mu(R) + \mu(I) = 2 > 0$, so Axiom III(a) (conservation) holds.

However, Axiom III(b) (projection–measure invariance) fails: taking $B = \{r\} \subseteq R$,

$$\Pi_R^{-1}(\{r\}) = X = \{r, i\}, \qquad \mu(\Pi_R^{-1}(\{r\})) = m(r) + m(i) = 2,$$

while $\mu(\{r\}) = 1$. Hence $\mu(\Pi_R^{-1}(B)) \neq \mu(B)$, and Axiom III(b) is violated. Consequently Axiom III as a whole fails. □

Remark 6.5 ($\mathcal{D}_{\neg\mathrm{III}}$ as the reason Axiom III(b) is needed in the reduction)**.** The datum $\mathcal{D}_{\neg\mathrm{III}}$ of Proposition 6.4 plays a second role beyond separating Axiom III: it witnesses the necessity of the Axiom III(b) hypothesis in part (iii) of Theorem 8.18. Explicitly, $\mathcal{D}_{\neg\mathrm{III}}$ satisfies Axioms I, II, III(a), III(c), fiber measurability, the regularity hypothesis (R–fin), and the full partition $X = R \sqcup I$; yet it is not admissible (because III(b) fails), while its restriction $\mathcal{D}_{\neg\mathrm{III}}|_R$ is the one–point identity–retraction datum $(\{r\}, \{r\})$ with total mass 1 and trivial relation $\Delta_{\{r\}}$, which is admissible. Hence without the Axiom III(b) hypothesis, admissibility of $\mathcal{D}|_R$ would not imply admissibility of $\mathcal{D}$.

6.2 Internal independence of Axiom III

Axiom III comprises three distinct assertions:

(a) *conservation*: $\mu(R) + \mu(I) = E_0 > 0$;

(b) *projection–measure invariance*: $\mu(\Pi_R^{-1}(B)) = \mu(B)$ for all measurable $B \subseteq R$;

(c) *coupling law*: $\mu^{\otimes 2}((B \times X) \cap G) = \mu(B) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G)$ for all $B \in \mathcal{A}$.

We show that these three subclauses are independent: no one of them follows from Axioms I, II together with the other two.

Proposition 6.6 (Internal independence of Axiom III)**.** *Subclauses* (a), (b), (c) *of Axiom III are mutually independent, relative to Axioms I and II.*

Proof. We exhibit three structures satisfying Axioms I, II and exactly two of (a), (b), (c).

(a) is independent. Let $X = \{r\}$, $R = \{r\}$, $I = \varnothing$, $\Pi_R = \mathrm{id}_X$, $G = \Delta_X$, and $\mu \equiv 0$ (the zero charge), with $\eta = 0$. Axioms I and II hold. Subclause (b) is trivial ($\Pi_R = \mathrm{id}$), and subclause (c) holds (both sides equal 0). However, $\mu(R) + \mu(I) = 0$, so no $E_0 > 0$ satisfies conservation; subclause (a) fails. This shows (a) is not derivable from Axioms I, II, (b), (c).

(b) is independent. The structure $\mathcal{D}_{\neg\mathrm{III}}$ of Proposition 6.4 satisfies Axioms I, II and subclause (a) (with $E_0 = 2$), and a direct check (as in the proof of Proposition 6.4) shows that the coupling law (c) holds with $\eta = 0$: for $B = \{r\}$, $\mu^{\otimes 2}((\{r\} \times X) \cap \Delta_X) = m(r)^2 = 1 = \mu(\{r\})$, and similarly for $B = \{i\}$ and $B = X$. However, subclause (b) fails as shown in Proposition 6.4. Hence (b) is not derivable from Axioms I, II, (a), (c).

(c) is independent. Let $X = \{a, b\}$, $R = X$, $I = \varnothing$, $\Pi_R = \mathrm{id}_X$, $G = X \times X$, with point weights $m(a) = 1$, $m(b) = 0$, and $\eta = 1/2$. Axioms I and II hold (the total relation $X \times X$ is reflexive, symmetric, and idempotent). Subclause (a): $\mu(R) + \mu(I) = 1 > 0$, take $E_0 = 1$. Subclause (b): $\Pi_R = \mathrm{id}_X$ makes invariance trivial. However, for $B = \{a\}$,

$$\mu^{\otimes 2}((\{a\} \times X) \cap (X \times X)) = m(a)\big(m(a) + m(b)\big) = 1 \cdot 1 = 1,$$

while

$$\mu(B) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G) = 1 + \tfrac{1}{2} \cdot 1 = \tfrac{3}{2} \neq 1,$$

so the coupling law fails. Hence (c) is not derivable from Axioms I, II, (a), (b). □

6.3 The main independence theorem and minimality corollary

Theorem 6.7 (Independence)**.** *The axioms* I, II, III *of Section 3 are mutually independent: for each* $\mathsf{X} \in \{\mathrm{I}, \mathrm{II}, \mathrm{III}\}$ *there exists a pre-structural datum over the ambient signature of Definition 3.1 satisfying all axioms except* X*. Moreover, the three subclauses* (a), (b), (c) *of Axiom III are themselves mutually independent, relative to Axioms I and II.*

Proof. Combine Propositions 6.1, 6.2, 6.4, and 6.6. □

Corollary 6.8 (Minimality)**.** *The axiom system* $\{\mathrm{I}, \mathrm{II}, \mathrm{III}\}$ *is minimal in the following sense: removing any one axiom, or any one subclause of Axiom III, yields a strictly weaker system admitting structures not admitted by the full system.*

Proof. Immediate from Theorem 6.7: the separating models of Propositions 6.1, 6.2, 6.4, 6.6 witness the claim in each case. □

Remark 6.9 (On the baseline analytic structure)**.** The independence results above are stated relative to the admitted analytic background: a set algebra $\mathcal{A} \subseteq \mathcal{P}(X)$, a finitely additive measure μ, a product charge $\mu^{\otimes 2}$ satisfying the rectangle rule, and a retraction

$\Pi_R : X \to R$ together with a binary relation $G \subseteq X \times X$. We do not claim purely logical independence stripped of this analytic setting; such a claim would require encoding the axioms in a formal system without those primitives, which is beyond the scope of this paper.

Remark 6.10 (Roles of the three axioms)**.** The three separating models clarify the distinct structural roles of the three axioms:

- Axiom I governs the *projective skeleton* of the datum (retraction data and measurability of preimages). Its failure in $\mathcal{D}_{\neg\mathrm{I}}$ is visible already at the level of Π_R, before measure-theoretic content is invoked.
- Axiom II governs the *closure structure* of the interaction relation. Its failure in $\mathcal{D}_{\neg\mathrm{II}}$ is set-theoretic ($G \circ G \supsetneq G$), independent of measure.
- Axiom III governs the *quantitative coupling* between measure, projection, and relation. Its three subclauses separate cleanly: conservation fixes total mass, invariance governs pointwise behavior under Π_R, and the coupling law governs the two-point curvature load.

7 σ–additive refinement and constraints

This paper is formulated for finitely additive measures in order to keep the axioms minimal and to ensure broad model existence. It is natural to ask what changes if one requires μ to be σ–additive (and hence admits the standard product measure by the classical construction).

7.1 Setup

Assume (X, Σ, μ) is a σ–additive measure space (see e.g. [5, 6]). In order to speak unambiguously about the product measure $\mu^{\otimes 2}$ on $(X \times X, \Sigma \otimes \Sigma)$ we assume μ is σ–finite when needed. In that case the standard product measure exists and is uniquely characterized by

$$\mu^{\otimes 2}(A \times B) = \mu(A)\mu(B) \qquad (A, B \in \Sigma),$$

together with σ–additivity on $\Sigma \otimes \Sigma$.

We emphasize that the conclusions below use only the $B = X$ instance of the coupling law, and thus apply in any σ–additive setting where $\mu^{\otimes 2}$ is defined.

7.2 A universal constraint from the coupling law

The coupling law in Axiom III implies a strong global constraint already at $B = X$.

Proposition 7.1 (Global constraint at $B = X$)**.** *Assume μ is σ–additive and $\mu^{\otimes 2}$ is the standard product measure. If the coupling law holds for $B = X$, then*

$$(1 - \eta)\, \mu^{\otimes 2}(G) = \mu(X).$$

In particular, if $\mu(X) \in (0, \infty)$ then

$$\mu^{\otimes 2}(G) = \frac{\mu(X)}{1 - \eta}.$$

Proof. Take $B = X$ in the coupling law:

$$\mu^{\otimes 2}(G) = \mu(X) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(X) \times X) \cap G).$$

Since $(X \times X) \cap G = G$ and $\Pi_R^{-1}(X) = X$, this becomes $\mu^{\otimes 2}(G) = \mu(X) + \eta\, \mu^{\otimes 2}(G)$, i.e. $(1 - \eta)\mu^{\otimes 2}(G) = \mu(X)$. □

Corollary 7.2 (Finite measure bound)**.** *Assume $\mu(X) = M \in (0, \infty)$. Then the coupling law at $B = X$ forces*

$$\frac{1}{1-\eta} \leq M.$$

Equivalently, $\eta \leq 1 - \frac{1}{M}$.

Proof. By Proposition 7.1, $\mu^{\otimes 2}(G) = M/(1-\eta)$. Since $\mu^{\otimes 2}(G) \leq \mu^{\otimes 2}(X \times X) = M^2$, we obtain $M/(1-\eta) \leq M^2$, i.e. $1/(1-\eta) \leq M$. □

Corollary 7.3 (Probability case forces $\eta = 0$)**.** *If μ is a probability measure, i.e. $\mu(X) = 1$, then the coupling law at $B = X$ implies*

$$\eta = 0 \quad \textit{and} \quad \mu^{\otimes 2}(G) = 1,$$

so G has full $\mu^{\otimes 2}$–measure in $X \times X$.

Proof. From Proposition 7.1, $\mu^{\otimes 2}(G) = 1/(1-\eta) \geq 1$. But $\mu^{\otimes 2}(G) \leq \mu^{\otimes 2}(X \times X) = 1$, hence $\mu^{\otimes 2}(G) = 1$ and $\eta = 0$. □

Proposition 7.4 (Feasibility window for $\eta \neq 0$ under finite total mass)**.** *Assume μ is σ–additive, σ–finite, and $\mu(X) = M \in (0, \infty)$. If the coupling law holds at $B = X$, then necessarily*

$$0 \leq \eta \leq 1 - \frac{1}{M},$$

and moreover $\mu^{\otimes 2}(G) = \frac{M}{1-\eta} \in (0, M^2]$. In particular, $\eta \neq 0$ is possible only when $M > 1$ (i.e. the measure is not probability-normalized).

Proof. The bound on η is Corollary 7.2. The identity for $\mu^{\otimes 2}(G)$ follows from Proposition 7.1. □

Remark 7.5 (Infinite-mass regimes)**.** If $\mu(X) = +\infty$, Proposition 7.1 becomes $+\infty = (1-\eta)\mu^{\otimes 2}(G)$, which does not by itself exclude $\eta \neq 0$. In such regimes one must use additional instances of the coupling law (i.e. varying B) or add normalization assumptions to obtain effective constraints.

Remark 7.6 (Scaling intuition)**.** In an approximating sequence of finite models $(X_n, \mu_n, G_n, \Pi_{R,n}, \eta_n)$, the limiting behavior of η_n is fundamentally normalization–dependent. Under standard probability normalization ($\mu_n(X_n) = 1$), the global constraint typically forces $\eta_n \to 0$, often rendering the coupling trivial. Conversely, in regimes where $\mu_n(X_n) = M_n \to M > 1$, η_n can stabilize to a nonzero constant compatible with the total mass M. In large–mass regimes ($M_n \to \infty$), the global identity does not by itself force $\eta_n \to 0$, shifting the structural burden to local instances of the coupling law (varying B), which become essential for characterizing the infinitesimal structure of the limit.

Further directions (minimal). (i) Study additional instances of the coupling law (varying B) to obtain nontrivial constraints beyond the $B = X$ global identity. (ii) Classify $\eta \neq 0$ solutions under normalization choices (finite M, infinite mass, or σ–finite restrictions) and determine when non-diagonal G must have full or intermediate $\mu^{\otimes 2}$–measure. (iii) Compare the finitely additive model families of Section 5 with σ–additive realizability, identifying the exact obstruction created by probability normalization.

Remark 7.7 (Interpretation)**.** Corollary 7.3 explains why nontrivial $\eta \neq 0$ families are naturally formulated either (i) in the finitely additive setting, or (ii) for σ–additive measures that are not normalized as probabilities. The global constraint arises solely from the $B = X$ instance of the coupling law; under a fixed normalization $\mu(X) = M$, the admissible range of η shrinks as M decreases, and any sequence of models in which η is to stabilize away from zero therefore requires M to grow accordingly.

8 Categorical structure

We record a basic categorical organization of structural models: the appropriate notion of morphism, the resulting category, and the preservation of the coupling law under morphisms.

8.1 Morphisms and the category Struct

Throughout this section we fix the scalar $\eta \in [0, 1)$, but treat $E_0 \in (0, \infty)$ as *object data*: it is recorded as part of each structural model and is allowed to vary between objects. Morphisms interact with E_0 via (M4) below, which yields an automatic inequality $E_0' \leq E_0$ (see Remark 8.2), rather than an a priori identification. From this point on, $\mathbf{Struct}_\eta$ denotes the category whose objects are admissible structural models (Definition 3.3) with common $\eta \in [0, 1)$, and whose morphisms are defined below.

Definition 8.1 (Morphism of structural models)**.** Let $\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ and $\mathcal{M}' = (X', \mathcal{A}', \mu', \mu'^{\otimes 2}, R', I', \Pi_{R'}, G', E_0', \eta)$ be admissible structural models (Definition 3.3) sharing a common value of $\eta \in [0, 1)$; the scalars E_0, E_0' are part of the object data and need not agree a priori. A *morphism* $\phi : \mathcal{M} \to \mathcal{M}'$ is a map $\phi : X \to X'$ satisfying:

(M1) Measurability: $\phi^{-1}(B') \in \mathcal{A}$ for every $B' \in \mathcal{A}'$, and $(\phi \times \phi)^{-1}(S') \in \mathcal{A} \otimes \mathcal{A}$ for every $S' \in \mathcal{A}' \otimes \mathcal{A}'$.

(M2) Projection commutativity: $\phi \circ \Pi_R = \Pi_{R'} \circ \phi$.

(M3) Relation preservation: $(\phi \times \phi)(G) \subseteq G'$, i.e., $(x, y) \in G$ implies $(\phi(x), \phi(y)) \in G'$.

(M4) Measure–preservation: $\mu(\phi^{-1}(B')) = \mu'(B')$ for every $B' \in \mathcal{A}'$, and $\mu^{\otimes 2}((\phi \times \phi)^{-1}(S')) = \mu'^{\otimes 2}(S')$ for every $S' \in \mathcal{A}' \otimes \mathcal{A}'$.

Remark 8.2 (E_0 under morphisms)**.** Axiom (M4) alone controls the relation between source and target scalars:

$$E_0' = \mu'(R') + \mu'(I') = \mu(\phi^{-1}(R')) + \mu(\phi^{-1}(I')) = \mu\big(\phi^{-1}(R' \cup I')\big) \leq \mu(X).$$

If $\mathcal{M}$ has full partition ($X = R \sqcup I$), then $\mu(X) = E_0$ and hence $E_0' \leq E_0$. Equality $E_0' = E_0$ holds when, in addition, $\phi^{-1}(R' \cup I') = X$, for instance when ϕ is surjective and $\mathcal{M}'$ also has full partition. Morphisms are thus in general mass–non–increasing, and mass–preserving in the natural full–partition surjective case. No separate scalar–identification axiom is imposed, since the relation between E_0 and E_0' is already controlled by (M4).

Remark 8.3 (Automatic consequences)**.** Three consequences follow directly from the definition:

- $\phi(R) \subseteq R'$, since $\phi(R) = \phi(\Pi_R(X)) = \Pi_{R'}(\phi(X)) \subseteq R'$ by (M2) and Axiom I for $\mathcal{M}'$.
- For $B' \subseteq R'$ measurable, $\mu'(B') = \mu(\phi^{-1}(B'))$, which combined with projection–measure invariance in $\mathcal{M}'$ recovers invariance in $\mathcal{M}$ along ϕ.

- The pushforward $\phi_*\mu^{\otimes 2}$ coincides with $\mu'^{\otimes 2}$ on rectangles, since

$$(\phi_*\mu^{\otimes 2})(A' \times B') = \mu^{\otimes 2}\big(\phi^{-1}(A') \times \phi^{-1}(B')\big) = \mu(\phi^{-1}(A'))\,\mu(\phi^{-1}(B'))$$

  which by (M4) equals $\mu'(A')\,\mu'(B') = \mu'^{\otimes 2}(A' \times B')$, using the rectangle rule.

Proposition 8.4 (Composition and identity)**.** *Morphisms of structural models compose: if $\phi : \mathcal{M} \to \mathcal{M}'$ and $\psi : \mathcal{M}' \to \mathcal{M}''$ are morphisms (sharing a common η), then $\psi \circ \phi : \mathcal{M} \to \mathcal{M}''$ is a morphism. Moreover, the identity map $\mathrm{id}_X : \mathcal{M} \to \mathcal{M}$ is a morphism. Hence structural models with a common value of $\eta \in [0, 1)$ (with E_0 allowed to vary as object data) together with their morphisms form a category, which we denote by* $\mathbf{Struct}_\eta$.

Proof. We check the four morphism axioms for $\psi \circ \phi$.

(M1) Measurability of $\psi \circ \phi$ follows from $(\psi \circ \phi)^{-1}(B'') = \phi^{-1}(\psi^{-1}(B'')) \in \mathcal{A}$ since $\psi^{-1}(B'') \in \mathcal{A}'$ and $\phi^{-1}(\mathcal{A}') \subseteq \mathcal{A}$. The product–measurability condition follows by the same argument applied to $(\psi \times \psi) \circ (\phi \times \phi)$.

(M2) Projection commutativity:

$$(\psi \circ \phi) \circ \Pi_R = \psi \circ (\phi \circ \Pi_R) = \psi \circ \Pi_{R'} \circ \phi = \Pi_{R''} \circ \psi \circ \phi = \Pi_{R''} \circ (\psi \circ \phi).$$

(M3) Relation preservation:

$$((\psi \circ \phi) \times (\psi \circ \phi))(G) = (\psi \times \psi)((\phi \times \phi)(G)) \subseteq (\psi \times \psi)(G') \subseteq G''.$$

(M4) Measure–preservation, for $B'' \in \mathcal{A}''$:

$$\mu((\psi \circ \phi)^{-1}(B'')) = \mu(\phi^{-1}(\psi^{-1}(B''))) = \mu'(\psi^{-1}(B'')) = \mu''(B'').$$

The product–measure identity follows analogously. Identity morphism is trivial. □

8.2 Preservation of the coupling law

Morphisms of structural models preserve not only the listed data but also the coupling law derived from Axiom III.

Proposition 8.5 (Coupling law is transported along exact pullback morphisms)**.** *Let $\phi : \mathcal{M} \to \mathcal{M}'$ be a morphism. Suppose $\mathcal{M}$ satisfies Axiom III (coupling law). Then:*

(i) (Source–side inequality) *Under the hypothesis $(\phi \times \phi)^{-1}(G') \subseteq G$, for every $B' \in \mathcal{A}'$,*

$$\mu'^{\otimes 2}\big((B' \times X') \cap G'\big) \;\le\; \mu^{\otimes 2}\big((\phi^{-1}(B') \times X) \cap G\big),$$

and the right–hand side admits the coupling–law expansion for $\mathcal{M}$:

$$\mu^{\otimes 2}\big((\phi^{-1}(B') \times X) \cap G\big) = \mu(\phi^{-1}(B')) + \eta\,\mu^{\otimes 2}\big((\Pi_R^{-1}(\phi^{-1}(B')) \times X) \cap G\big).$$

(ii) (Equality under exact pullback and surjectivity) *If in addition ϕ is surjective and $(\phi \times \phi)^{-1}(G') = G$, then $\mathcal{M}'$ satisfies the coupling law in full:*

$$\mu'^{\otimes 2}\big((B' \times X') \cap G'\big) = \mu'(B') + \eta\,\mu'^{\otimes 2}\big((\Pi_{R'}^{-1}(B') \times X') \cap G'\big) \qquad (\forall B' \in \mathcal{A}').$$

Proof. Let $B' \in \mathcal{A}'$. By (M4),

$$\mu'^{\otimes 2}((B' \times X') \cap G') = \mu^{\otimes 2}\big((\phi \times \phi)^{-1}((B' \times X') \cap G')\big).$$

Since $(\phi \times \phi)^{-1}(B' \times X') = \phi^{-1}(B') \times X$, we have

$$(\phi \times \phi)^{-1}((B' \times X') \cap G') = (\phi^{-1}(B') \times X) \cap (\phi \times \phi)^{-1}(G').$$

Part (i). Under $(\phi\times\phi)^{-1}(G') \subseteq G$, the set on the right is contained in $(\phi^{-1}(B')\times X)\cap G$; by monotonicity of $\mu^{\otimes 2}$,

$$\mu'^{\otimes 2}((B' \times X') \cap G') \leq \mu^{\otimes 2}((\phi^{-1}(B') \times X) \cap G).$$

By the coupling law for $\mathcal{M}$ applied to $B = \phi^{-1}(B')$,

$$\mu^{\otimes 2}((\phi^{-1}(B') \times X) \cap G) = \mu(\phi^{-1}(B')) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(\phi^{-1}(B')) \times X) \cap G),$$

giving the inequality asserted in (i), with the right–hand side admitting the stated expansion.

Part (ii). Under the stronger hypotheses $(\phi \times \phi)^{-1}(G') = G$ and ϕ surjective, the inclusion in the displayed identity becomes equality:

$$(\phi \times \phi)^{-1}((B' \times X') \cap G') = (\phi^{-1}(B') \times X) \cap G,$$

and hence

$$\mu'^{\otimes 2}((B' \times X') \cap G') = \mu^{\otimes 2}((\phi^{-1}(B') \times X) \cap G).$$

Applying the coupling law for $\mathcal{M}$ to $B = \phi^{-1}(B')$, using (M2) to rewrite

$$\Pi_R^{-1}(\phi^{-1}(B')) = \phi^{-1}(\Pi_{R'}^{-1}(B')),$$

and applying (M4) for both μ and $\mu^{\otimes 2}$:

$$\begin{aligned}\mu'^{\otimes 2}((B' \times X') \cap G') &= \mu(\phi^{-1}(B')) + \eta\, \mu^{\otimes 2}((\phi^{-1}(\Pi_{R'}^{-1}(B')) \times X) \cap G) \\ &= \mu'(B') + \eta\, \mu'^{\otimes 2}((\Pi_{R'}^{-1}(B') \times X') \cap G'),\end{aligned}$$

which is the coupling law in $\mathcal{M}'$, valid for every $B' \in \mathcal{A}'$ (using surjectivity of ϕ to range over all B'). □

Remark 8.6 (On the hypotheses)**.** Part (ii) of Proposition 8.5 is the intended "transport of coupling law" statement; the strong hypothesis $(\phi \times \phi)^{-1}(G') = G$ combined with surjectivity of ϕ is automatic, for instance, when $\mathcal{M}$ is the full pullback of $\mathcal{M}'$ along ϕ (i.e., when $\mathcal{M}$ is obtained from $\mathcal{M}'$ by pulling back the measure and relation along ϕ). In the weaker inclusion case of part (i), only an inequality survives, and the coupling law of $\mathcal{M}'$ is bounded by the coupling–law expression of $\mathcal{M}$ on the pulled–back test set. The discrepancy between (i) and (ii) records precisely the information lost when ϕ fails to be surjective or the relation is preserved only inclusively.

8.3 Idempotent endomorphisms in Struct

The axioms admit a natural categorical reformulation in terms of idempotents.

Proposition 8.7 (Retraction as a split idempotent on objects)**.** *Let $\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ be a structural model. The map $\Pi_R : X \to X$ is idempotent ($\Pi_R \circ \Pi_R = \Pi_R$) at the level of the underlying measurable-relation data. Restricting codomain to $R \in \mathcal{A}$, Π_R factors as*

$$\Pi_R \;=\; \iota_R \circ \rho_R,$$

where $\rho_R : X \to R$ is the surjection defined by Π_R and $\iota_R : R \hookrightarrow X$ is the inclusion. The equality $\rho_R \circ \iota_R = \mathrm{id}_R$ holds by Axiom I.

Proof. Idempotence is Axiom I. The factorization $\Pi_R = \iota_R \circ \rho_R$ is definitional: ρ_R is simply Π_R with codomain restricted to R, and $\iota_R : R \to X$ is the set–theoretic inclusion. The identity $\rho_R \circ \iota_R = \mathrm{id}_R$ follows from $\Pi_R|_R = \mathrm{id}_R$ (Axiom I). □

Proposition 8.8 (Relation idempotence at the morphism level)**.** *The relation $G \subseteq X \times X$, viewed as a binary endorelation, is an idempotent endomorphism in the poset category of binary relations on X under relational composition: $G \circ G = G$ (Axiom II). Consequently G defines an equivalence relation $\sim_G$ on X.*

Proof. Idempotence is Axiom II; Proposition 4.1(iii) upgrades reflexive, symmetric, idempotent relations to equivalence relations. □

Remark 8.9 (Monad–comonad perspective)**.** Propositions 8.7 and 8.8 together place the two structural idempotents of a structural model in analogy with the two sides of an idempotent monad–comonad pair:

- The retraction Π_R plays the role of the monad component: it is an idempotent endomorphism with a distinguished inclusion section ι_R.
- The relation G plays the role of the comonad component: its idempotence generates equivalence classes that organize X into a quotient structure.

The coupling law of Axiom III then expresses a quantitative compatibility between these two idempotent structures, mediated by the measure μ and the parameter η.

8.4 The subcategory of $\Pi_R = \mathrm{id}$ models

Within $\mathbf{Struct}_\eta$, the condition $\Pi_R = \mathrm{id}_X$ singles out a structurally simpler subcategory.

Definition 8.10 (Subcategory of identity–retraction models)**.** Let $\mathbf{Struct}_\eta^{\mathrm{id}} \subseteq \mathbf{Struct}_\eta$ be the full subcategory consisting of models $\mathcal{M}$ with $\Pi_R = \mathrm{id}_X$ (so $R = X$, $I = \varnothing$).

Proposition 8.11 (Restriction functor, under fiber hypotheses)**.** *Let $\mathbf{Struct}_\eta^{\mathrm{fib}} \subseteq \mathbf{Struct}_\eta$ denote the full subcategory whose objects are those structural models $\mathcal{M}$ satisfying the measurability hypothesis of Proposition 8.15 (namely $\{r\} \in \mathcal{A}$ and $F_r = \Pi_R^{-1}(\{r\}) \in \mathcal{A}$ for every $r \in R$) together with one of the regularity hypotheses* (R–fin) *or* (R–ctbl) *of that proposition. Then the assignment*

$$\rho : \mathbf{Struct}_\eta^{\mathrm{fib}} \longrightarrow \mathbf{Struct}_\eta^{\mathrm{id}}, \qquad \mathcal{M} \mapsto \mathcal{M}|_R, \quad \phi \mapsto \phi|_R,$$

defines a functor, where $\mathcal{M}|_R$ is the model with underlying set R, set algebra $\mathcal{A} \cap \mathcal{P}(R)$, measure $\mu|_R$, product charge $\mu^{\otimes 2}|_{R\times R}$, trivial retraction $\Pi_{R|_R} = \mathrm{id}_R$, restricted relation $G|_R := G \cap (R \times R)$, scalar $E_0(\mathcal{M}|_R) := \mu(R)$, and coupling parameter η.

Proof. We verify that $\mathcal{M}|_R$ satisfies Axioms I–III, so that $\mathcal{M}|_R \in \mathbf{Struct}_\eta^{\mathrm{id}}$. The argument relies only on Proposition 8.15 (the fiber–annihilation consequence $\mu(X \setminus R) = 0$ under the hypotheses defining $\mathbf{Struct}_\eta^{\mathrm{fib}}$) and on Axioms I–III for $\mathcal{M}$; it does not invoke Theorem 8.18.

Axiom I on $\mathcal{M}|_R$. Immediate from $\Pi_{R|_R} = \mathrm{id}_R$: the retraction is idempotent, and $\Pi_{R|_R}|_R = \mathrm{id}_R$ trivially, with measurability of preimages holding on $\mathcal{A} \cap \mathcal{P}(R)$.

Axiom II on $\mathcal{M}|_R$. The restricted relation $G|_R := G \cap (R \times R)$ is reflexive on R (since $\Delta_R \subseteq \Delta_X \subseteq G$ and $\Delta_R \subseteq R \times R$) and symmetric by restriction. For idempotence, observe first that $G|_R \circ G|_R \subseteq G \circ G \cap (R \times R) = G \cap (R \times R) = G|_R$ (using Axiom II for $\mathcal{M}$ and the fact that composing relations on $R \times R$ stays in $R \times R$). Conversely, if $(x, z) \in G|_R$, then $x, z \in R$ and, by reflexivity of $G|_R$ giving $(x, x) \in G|_R$, we have $(x, z) = (x, x) \circ (x, z) \in G|_R \circ G|_R$. Therefore $G|_R \circ G|_R = G|_R$.

Axiom III(a) on $\mathcal{M}|_R$. We must show $\mu(R) > 0$. By Proposition 8.15, $\mu(X \setminus R) = 0$. Since $I \subseteq X \setminus R$ by construction, $\mu(I) = 0$. Axiom III(a) for $\mathcal{M}$ gives $\mu(R) + \mu(I) = E_0(\mathcal{M}) > 0$, hence $\mu(R) = E_0(\mathcal{M}) > 0$. We record this positive scalar as $E_0(\mathcal{M}|_R) := \mu(R)$.

Axiom III(b) on $\mathcal{M}|_R$. Trivial, since $\Pi_{R|_R} = \mathrm{id}_R$.

Axiom III(c) on $\mathcal{M}|_R$. We show directly that for every $B \in \mathcal{A} \cap \mathcal{P}(R)$,

$$\mu^{\otimes 2}((B \times R) \cap G|_R) = \mu(B) + \eta\, \mu^{\otimes 2}((B \times R) \cap G|_R),$$

equivalently $(1-\eta)\mu^{\otimes 2}((B \times R) \cap G|_R) = \mu(B)$. Fix $B \in \mathcal{A} \cap \mathcal{P}(R)$, so $B \subseteq R$ and hence $\Pi_R^{-1}(B) \cap R = B$ (by $\Pi_R|_R = \mathrm{id}_R$). Applying the coupling law for $\mathcal{M}$ to this B (viewed as an element of $\mathcal{A}$):

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu(B) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G). \tag{$*$}$$

Since $\mu(X \setminus R) = 0$, monotonicity and the rectangle rule give

$$S \subseteq X \times (X \setminus R) \text{ measurable} \implies \mu^{\otimes 2}(S) \le \mu^{\otimes 2}(X \times (X \setminus R)) = \mu(X)\,\mu(X \setminus R) = 0,$$

and similarly $\mu^{\otimes 2}(S') = 0$ for every measurable $S' \subseteq (X \setminus R) \times X$. Consequently,

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu^{\otimes 2}((B \times R) \cap G) = \mu^{\otimes 2}((B \times R) \cap G|_R),$$

where the last equality uses $B \subseteq R$ so $(B \times R) \cap G = (B \times R) \cap G \cap (R \times R) = (B \times R) \cap G|_R$. For the right–hand side of $(*)$, note that $\Pi_R^{-1}(B) \setminus B \subseteq X \setminus R$: indeed, if $x \in \Pi_R^{-1}(B) \cap R$ then $\Pi_R(x) = x \in B$ (Axiom I), so $x \in B$; the contrapositive gives $\Pi_R^{-1}(B) \setminus B \subseteq X \setminus R$. Hence

$$\mu^{\otimes 2}((\Pi_R^{-1}(B) \times X) \cap G) = \mu^{\otimes 2}((B \times X) \cap G) = \mu^{\otimes 2}((B \times R) \cap G|_R),$$

using that the "extra" set $(\Pi_R^{-1}(B) \setminus B) \times X$ is contained in $(X \setminus R) \times X$, hence $\mu^{\otimes 2}$–null by monotonicity as above, and applying the previous reduction. Substituting both sides of $(*)$:

$$\mu^{\otimes 2}((B \times R) \cap G|_R) = \mu(B) + \eta\, \mu^{\otimes 2}((B \times R) \cap G|_R),$$

which is Axiom III(c) for $\mathcal{M}|_R$.

Functoriality. If $\phi : \mathcal{M} \to \mathcal{M}'$ is a morphism in $\mathbf{Struct}_\eta^{\mathrm{fib}}$, then $\phi(R) \subseteq R'$ by Remark 8.3, so $\phi|_R : R \to R'$ is well–defined. Axioms (M1)–(M4) for $\phi|_R$ follow by restriction from those of ϕ. Composition and identity are preserved, so ρ is a functor. □

Remark 8.12 (The restriction functor as a retraction of categories)**.** The subcategory $\mathbf{Struct}_\eta^{\mathrm{id}}$ includes canonically into $\mathbf{Struct}_\eta^{\mathrm{fib}}$ via the inclusion ι that regards an identity–retraction object as already satisfying the fiber hypotheses trivially (with $F_r = \{r\}$ for every $r \in R = X$). The composition $\rho \circ \iota$ is the identity on $\mathbf{Struct}_\eta^{\mathrm{id}}$: restricting an identity–retraction model to $R = X$ returns the model unchanged. In this precise sense, ρ is a retraction of $\mathbf{Struct}_\eta^{\mathrm{fib}}$ onto its subcategory $\mathbf{Struct}_\eta^{\mathrm{id}}$, mirroring at the categorical level the object–level retraction $\Pi_R : X \to R$.

Remark 8.13 (Connection to the classification theorem)**.** Within the full subcategory of $\mathbf{Struct}_\eta^{\mathrm{id}}$ whose objects satisfy the full hypotheses of Theorem 5.4, that theorem provides a dichotomy: for every G–equivalence class C_k, either $\mu(C_k) = 0$ or $\mu(C_k) = (1-\eta)^{-1}$. Equivalently, every positive–mass class satisfies $\mu(C_k) = (1-\eta)^{-1}$. On that subcategory, each object can be described combinatorially by a partition $(X, (C_k)_{k \in K})$ together with admissible mass data on the classes (namely the dichotomy $\mu(C_k) \in \{0, (1-\eta)^{-1}\}$) and the induced product charge on $X \times X$; morphisms preserve the partition structure and the measure/charge data in the sense of Definition 8.1. The restriction functor ρ of Proposition 8.11 extracts this combinatorial core from an arbitrary structural model under the fiber hypotheses, provided the core satisfies those further hypotheses.

8.5 Quotient factorization and structural reduction

We now sharpen the restriction functor of Proposition 8.11 into a structural reduction theorem: under fiber measurability and regularity hypotheses, all curvature–load information relevant to Axioms I–III reduces to the identity–retraction core $\mathcal{M}|_R$, and the coupling law on $\mathcal{M}$ is equivalent to the coupling law on $\mathcal{M}|_R$. This extends the classification theorem (Theorem 5.4), which addresses the case $\Pi_R = \mathrm{id}_X$, to the general case.

Throughout this subsection let

$$\mathcal{D} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

be a pre-structural datum (Definition 3.1) with $\eta \in [0, 1)$ and $\mu(X) < \infty$. Additional assumptions (fiber measurability, regularity, partial axioms, or full admissibility) will be imposed explicitly in each statement.

Definition 8.14 (Fiber partition)**.** For each $r \in R$, the *fiber of* Π_R *over* r is

$$F_r := \Pi_R^{-1}(\{r\}) = \{x \in X : \Pi_R(x) = r\}.$$

Since Π_R is a function into R, the fibers $\{F_r\}_{r \in R}$ always partition X as a disjoint union indexed by the image of Π_R; if moreover Axiom I holds, then $\Pi_R|_R = \mathrm{id}_R$ so $r \in F_r$ for every $r \in R$, and $X = \bigsqcup_{r \in R} F_r$.

The fiber decomposition is the set–theoretic quotient of X under the equivalence relation $x \sim_\Pi y \iff \Pi_R(x) = \Pi_R(y)$; if Axiom I holds, this quotient is canonically identified with R via $r \leftrightarrow F_r$.

Proposition 8.15 (Fiber–measure annihilation)**.** *Assume Axiom I holds. Assume moreover that $\{r\} \in \mathcal{A}$ and $F_r \in \mathcal{A}$ for every $r \in R$, and that one of the following global regularity hypotheses holds:*

(R–fin) *R is finite; or*

(R–ctbl) *R is countable and μ is σ–additive on $\mathcal{A}$ (so $\mathcal{A}$ is assumed to be a σ–algebra).*

If Axiom III(b) (projection–measure invariance) holds, then

$$\mu(F_r \setminus \{r\}) = 0 \qquad (\forall\, r \in R),$$

and consequently $\mu(X \setminus R) = 0$: the measure μ is concentrated on the representative sector R.

Proof. By Axiom I, $\Pi_R|_R = \mathrm{id}_R$, so $\Pi_R(r) = r$ and $r \in F_r = \Pi_R^{-1}(\{r\})$ for every $r \in R$. Hence

$$F_r = \{r\} \sqcup (F_r \setminus \{r\})$$

is a disjoint union of sets in $\mathcal{A}$. By invariance (Axiom III(b)) applied to $B = \{r\} \subseteq R$,

$$\mu(\Pi_R^{-1}(\{r\})) = \mu(\{r\}),$$

so finite additivity gives

$$\mu(\{r\}) + \mu(F_r \setminus \{r\}) = \mu(\{r\}),$$

hence $\mu(F_r \setminus \{r\}) = 0$.

For the global claim $\mu(X \setminus R) = 0$, decompose

$$X \setminus R = \bigsqcup_{r \in R} (F_r \setminus \{r\}).$$

Under (R–fin) this is a finite disjoint union of null sets, and finite additivity gives $\mu(X \setminus R) = 0$. Under (R–ctbl) the union is countable, and σ–additivity of μ applied to the countable disjoint union gives again $\mu(X \setminus R) = 0$. □

Remark 8.16 (On the regularity hypothesis)**.** Without either hypothesis (R–fin) or (R–ctbl), the fiber–wise conclusion $\mu(F_r \setminus \{r\}) = 0$ still holds for each individual $r \in R$, but the global conclusion $\mu(X \setminus R) = 0$ may fail, since finite additivity alone does not control infinitely many null sets. A finitely additive measure on an uncountable R may assign positive total mass to an uncountable disjoint union of null sets. The hypotheses (R–fin), (R–ctbl) are the natural setting for the structural reduction theorem below; extensions to the uncountable σ–additive case require additional assumptions (e.g., separability of $\mathcal{A}$) which we do not pursue.

Remark 8.17 (Relation to Proposition 4.3)**.** Proposition 8.15 strengthens Proposition 4.3 (support concentration under full partition): the full partition hypothesis $X = R \sqcup I$ is replaced by the weaker measurability condition on fibers. The conclusion $\mu(X \setminus R) = 0$ now holds for any structural model whose fibers are measurable, regardless of whether $X = R \sqcup I$.

Theorem 8.18 (Quotient factorization)**.** *Let*

$$\mathcal{D} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

be a pre-structural datum (Definition 3.1) with $\eta \in [0, 1)$ and $\mu(X) < \infty$, satisfying the measurability hypothesis of Proposition 8.15 and one of the regularity hypotheses (R–fin) *or* (R–ctbl) *of that proposition. Let $\mathcal{D}|_R$ denote the restricted datum on R (with underlying set R, algebra $\mathcal{A} \cap \mathcal{P}(R)$, measure $\mu|_R$, restricted product charge, trivial retraction id_R, restricted relation $G|_R := G \cap (R \times R)$, scalar $\mu(R)$, and coupling parameter η).*

*(i) (*curvature load reduces to R*) If $\mathcal{D}$ satisfies Axiom I and Axiom III(b), then $\mu(X \setminus R) = 0$, and for all $B \in \mathcal{A}$, writing $B_R := B \cap R$,*

$$\mu^{\otimes 2}\big((B \times X) \cap G\big) = \mu^{\otimes 2}\big((B_R \times R) \cap (G|_R)\big).$$

*(ii) (*coupling laws are equivalent*) If $\mathcal{D}$ satisfies Axiom I and Axiom III(b), then the coupling law (Axiom III(c)) holds on $\mathcal{D}$ if and only if it holds on $\mathcal{D}|_R$.*

*(iii) (*admissibility reduces*) If $\mathcal{D}$ satisfies Axioms I, II, III(a), and III(b) as well as the full partition condition $X = R \sqcup I$, then $\mathcal{D}$ is an admissible structural model (i.e., an object of $\mathbf{Struct}_\eta$) if and only if $\mathcal{D}|_R$ is an admissible structural model (i.e., an object of $\mathbf{Struct}_\eta^{\mathrm{id}}$). In particular, whenever the restricted core $\mathcal{D}|_R$ also satisfies the full hypotheses of Theorem 5.4, the classification of admissible structural models reduces to the identity–retraction classification of Theorem 5.4. (The hypothesis "Axiom III(b)" is not redundant: the datum $\mathcal{D}_{\neg\mathrm{III}}$ of Proposition 6.4 satisfies Axioms I, II, III(a), III(c), fiber measurability, (R–fin), and the full partition $X = R \sqcup I$, yet fails Axiom III(b); its restriction $\mathcal{D}_{\neg\mathrm{III}}|_R$ is admissible while $\mathcal{D}_{\neg\mathrm{III}}$ itself is not, so the equivalence fails without III(b).)*

Proof. (i) By Proposition 8.15, which applies since $\mathcal{D}$ satisfies Axiom I and Axiom III(b) together with the measurability and regularity hypotheses, $\mu(X \setminus R) = 0$. Since $\mu(X \setminus R) = 0$, monotonicity and the rectangle rule give

$$\mu^{\otimes 2}(S) \;\le\; \mu^{\otimes 2}\big((X \setminus R) \times X\big) = \mu(X \setminus R)\,\mu(X) = 0$$

for every measurable $S \subseteq (X \setminus R) \times X$. Similarly, $\mu^{\otimes 2}(S') = 0$ for every measurable $S' \subseteq X \times (X \setminus R)$.

Decomposing $B = B_R \sqcup (B \setminus R)$ with $B_R := B \cap R$ (no full–partition hypothesis assumed) and using $R \sqcup (X \setminus R) = X$:

$$(B \times X) \cap G = \big((B_R \times R) \cap G\big) \sqcup \big((B_R \times (X \setminus R)) \cap G\big) \sqcup \big(((B \setminus R) \times X) \cap G\big).$$

The second term lies in $X \times (X \setminus R)$, hence has $\mu^{\otimes 2}$–measure zero. The third term lies in $(X \setminus R) \times X$ (since $B \setminus R \subseteq X \setminus R$), hence also has $\mu^{\otimes 2}$–measure zero. Therefore

$$\mu^{\otimes 2}((B \times X) \cap G) = \mu^{\otimes 2}((B_R \times R) \cap G) = \mu^{\otimes 2}((B_R \times R) \cap G|_R),$$

where the last equality uses $(B_R \times R) \cap G = (B_R \times R) \cap (G \cap (R \times R)) = (B_R \times R) \cap G|_R$, valid because $B_R \subseteq R$.

(ii) Assume first that the coupling law (Axiom III(c)) holds on $\mathcal{D}$. For $B \in \mathcal{A}$ with $B_R = B \cap R$, by (i) and Remark 3.2 ($\Pi_R^{-1}(B) = \Pi_R^{-1}(B_R)$):

$$\mu^{\otimes 2}((B_R \times R) \cap G|_R) = \mu(B) + \eta\, \mu^{\otimes 2}((\Pi_R^{-1}(B_R) \times R) \cap G|_R).$$

Since $\mu(X \setminus R) = 0$ we have $\mu(B) = \mu(B_R)$. Within $\mathcal{D}|_R$, the retraction is id_R, so $\Pi_R^{-1}(B_R) \cap R = B_R$. Substituting:

$$\mu^{\otimes 2}((B_R \times R) \cap G|_R) = \mu(B_R) + \eta\, \mu^{\otimes 2}((B_R \times R) \cap G|_R),$$

which is the coupling law on $\mathcal{D}|_R$ applied to B_R. As B ranges over $\mathcal{A}$, B_R ranges over $\mathcal{A} \cap \mathcal{P}(R)$, so the coupling law on $\mathcal{D}|_R$ follows. The converse runs the same argument in reverse: given the coupling law on $\mathcal{D}|_R$, identity (i) lifts it to $\mathcal{D}$.

(iii) We show the equivalence of admissibility. Axioms I and II for $\mathcal{D}$ imply Axioms I and II for $\mathcal{D}|_R$ (with retraction id_R and $G|_R$ inheriting reflexivity, symmetry, idempotence from G; see also the explicit argument for $G \circ G = G$ restricted to $R \times R$ given in the proof of Proposition 8.11). For Axiom III(a) on $\mathcal{D}|_R$: since $I \subseteq X \setminus R$ and $\mu(X \setminus R) = 0$ by (i), we have $\mu(I) = 0$. Axiom III(a) on $\mathcal{D}$ gives $\mu(R) + \mu(I) = E_0 > 0$, hence $\mu(R) = E_0 > 0$, which is precisely Axiom III(a) on $\mathcal{D}|_R$ with scalar $\mu(R)$. Axiom III(b) on $\mathcal{D}|_R$ is trivial because the retraction is id_R. Axiom III(c) on $\mathcal{D}|_R$ is equivalent to Axiom III(c) on $\mathcal{D}$ by part (ii). Combining, $\mathcal{D}$ satisfies Axioms I–III if and only if $\mathcal{D}|_R$ does, i.e., $\mathcal{D}$ is an admissible structural model if and only if $\mathcal{D}|_R$ is. This proves the equivalence of admissibility. The final classification claim (reduction to the identity–retraction classification of Theorem 5.4) follows by applying Theorem 5.4 to $\mathcal{D}|_R$ under its full hypotheses. □

Corollary 8.19 (Closed form of the curvature load)**.** *Under the hypotheses of Theorem 8.18, assume moreover that $\mathcal{D}$ is admissible and that the restricted core $\mathcal{D}|_R$ satisfies the full hypotheses of Theorem 5.4. Then the observed curvature load on any measurable $B \subseteq X$ is given by*

$$\mu^{\otimes 2}((B \times X) \cap G) = \sum_{k \in K} \mu(B_R \cap C_k)\, \mu(C_k),$$

where $\{C_k\}_{k \in K}$ are the $G|_R$–equivalence classes of the identity–retraction core $\mathcal{D}|_R$, and $\mu(C_k) \in \{0, (1-\eta)^{-1}\}$ for each class, with $\mu(C_k) = (1-\eta)^{-1}$ on positive–mass classes.

Proof. Combine Theorem 8.18(i) with Theorem 5.4, applied to the admissible core $\mathcal{D}|_R$ under its full hypotheses. □

Remark 8.20 (Structure of the reduction)**.** Theorem 8.18(iii) exhibits the pair $(\mathbf{Struct}_\eta^{\mathrm{id}}, \mathbf{Struct}_\eta)$ as analogous to a space and a null–set extension of it: passing from $\mathcal{D}|_R$ to $\mathcal{D}$ adds only measure–zero fiber data, invisible to the coupling law and the curvature load. This parallels the measure–theoretic fact that two measures agreeing off a null set are operationally indistinguishable.

Remark 8.21 (Place of the quotient factorization in the paper)**.** Theorem 8.18 bridges three earlier structural results into a single statement:

- It generalizes Proposition 4.3 (support concentration), extending $\mu(I) = 0$ from the full partition case to the fiber–measurability case.
- It promotes the restriction functor ρ of Proposition 8.11 from a categorical operation to a structural equivalence on curvature loads.
- It reduces the general classification problem to Theorem 5.4, under measurability hypotheses on fibers and the full hypotheses of Theorem 5.4 on the restricted core.

9 Examples

We conclude with two finite examples.

9.1 Finite representative/hidden pattern

Example 9.1 (Finite representative/hidden pattern)**.** Let X be a finite nonempty set, partitioned into nonempty blocks $(C_j)_{j\in J}$. For each j, choose a distinguished element $r_j \in C_j$ and define

$$R := \{r_j : j \in J\}, \qquad I := X \setminus R.$$

One may view R as the observable representative sector and I as a hidden sector; we define $\Pi_R : X \to R$ by $\Pi_R(x) = r_j$ whenever $x \in C_j$. Let $\mathcal{A} = \mathcal{P}(X)$ and choose weights $w_j \in \{0, 1\}$ with $\sum_j w_j > 0$. Define $\mu(B) = \sum_{j:r_j\in B} w_j$, and let

$$G := \bigcup_{j\in J} (C_j \times C_j)$$

be the union of equivalence relations on each block. Set $E_0 = \mu(X)$ and $\eta = 0$. Then Proposition 5.10 shows that

$$\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$$

is a structural model. Each block C_j represents a cluster of points sharing the same hidden content, with r_j as the open representative.

9.2 Abstract equivalence/bridge pattern

Example 9.2 (Abstract equivalence/bridge pattern)**.** Let X be any nonempty set equipped with an equivalence relation $\approx$. Assume $X/\approx$ is finite, and choose a representative r_j for each class C_j. Set

$$R := \{r_j\}, \qquad I := X \setminus R, \qquad G := \bigcup_j (C_j \times C_j),$$

and let $\Pi_R : X \to R$ send each $x \in C_j$ to r_j. Let $\mathcal{A} = \mathcal{P}(X)$. Choose weights $w_j \in \{0, 1\}$ with $\sum_j w_j > 0$ and define

$$\mu(B) = \sum_{j:r_j\in B} w_j.$$

Then by Proposition 5.10, the tuple $\mathcal{M} = (X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ with $E_0 = \mu(X)$ and $\eta = 0$ is a structural model. Heuristically, G encodes a bridge relation between configurations, grouping those that share the same structural type. The set R collects canonical representatives of these types, and Π_R retracts the full space X onto the simpler representative space R while preserving the chosen measure on R. The axioms formalize this pattern in a ZFC–internal way.

10 Conclusion

We have set up a minimal ZFC–internal axiom system for admissible structural models pre-structural data $(X, \mathcal{A}, \mu, \mu^{\otimes 2}, R, I, \Pi_R, G, E_0, \eta)$ satisfying Axioms I, II, and III and studied its basic properties.

The axiom system is consistent in ZFC (Section 5), and its three axioms, together with the three subclauses of Axiom III, are mutually independent (Theorem 6.7, Corollary 6.8). The coupling law of Axiom III admits an operator–theoretic reformulation as the unique bounded finitely additive solution of the Banach–contraction equation $f = T_\eta f$, with closed form $f_*(B) = \mu(B) + \frac{\eta}{1-\eta}\mu(\Pi_R^{-1}(B))$ (Theorem 4.8). Admissible structural models with a common η form a category $\mathbf{Struct}_\eta$ (Section 8); under fiber measurability and either (R–fin) or (R–ctbl), the general admissibility problem reduces to the identity–retraction case (Theorem 8.18), where the classification of Theorem 5.4 yields the block dichotomy $\mu(C_k) \in \{0, (1-\eta)^{-1}\}$. Under σ–additive probability normalization, the $B = X$ instance of the coupling law forces $\eta = 0$ (Section 7), so $\eta \neq 0$ families live in the finitely additive setting or among non–probability σ–additive measures.

Further directions. Several extensions suggest themselves: replacing $\mathcal{A}$ by a σ–algebra under hypotheses giving a well–defined product charge; enriching G to a weighted kernel $K : X \times X \to [0, \infty)$ and comparing with graph limits [15]; further categorical development, in particular an adjoint characterization of the restriction $\rho : \mathbf{Struct}_\eta^{\mathrm{fib}} \to \mathbf{Struct}_\eta^{\mathrm{id}}$ [17, 20]; operator–theoretic variants replacing μ by a state or trace; and scaling limits $X_n \to X$ controlled by local instances of the coupling law. Removing the fiber hypotheses in the quotient–factorization reduction, for instance in the uncountable σ–additive case, also requires further assumptions such as separability of $\mathcal{A}$.

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